



Final Design Project:
Gas Turbofan Engine Compressor

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ME 407: Introduction to Computational Fluid Dynamics (CFD)

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Spring 2024

The Cooper Union for the Advancement of Science and Art

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1. Problem Statement

This project aims to design and optimize the compressor of a new gas turbofan engine to minimize power consumption and maximize efficiency under given design requirements. The compressor stage is responsible for increasing the density of air before it is fed into a combustor and turbine. Computational Fluid Dynamics (CFD) will be utilized to simulate and find the optimal geometry of the compressor rotors and rotor speed.

2. Design Concept

According to research done by the National Aeronautics and Space Administration (NASA), the NACA 65-series airfoils have proved for their optimal use for compressor blades and thus have been widely used for high-performance, flight-type compressors¹. Among NACA 65-series airfoils, the 65-210 airfoil was chosen (shown in Figure 1) due to its simplicity of the geometry for analysis. In addition to its near symmetrical geometry, the maximum camber ratio at 50% chord is 1.1%, which falls under the range of practical application (1 to 3 %). Therefore, the 65-210 airfoil is suitable for this application.



Figure 1. NACA 65-210 Airfoil

(Source: Airfoil Tools)

As shown in Figure 2, the compressor consists of four main parts: a nose cone, stators and rotors, and wall. The nose cone helps streamline, reducing drag and ensuring smooth flow of air into the compressor. The hub diameter of the stage increases with a constant rate up to tenth stage while the blade length decreases with a constant rate to keep the maximum outlet diameter requirement shown in Table 1. After the tenth stage, the hub diameter is fixed while the blade length keeps decreasing, until the hub diameter is restricted by the minimum outlet diameter requirement. Figure 3 well shows the tapered stages.

¹ (Wright)

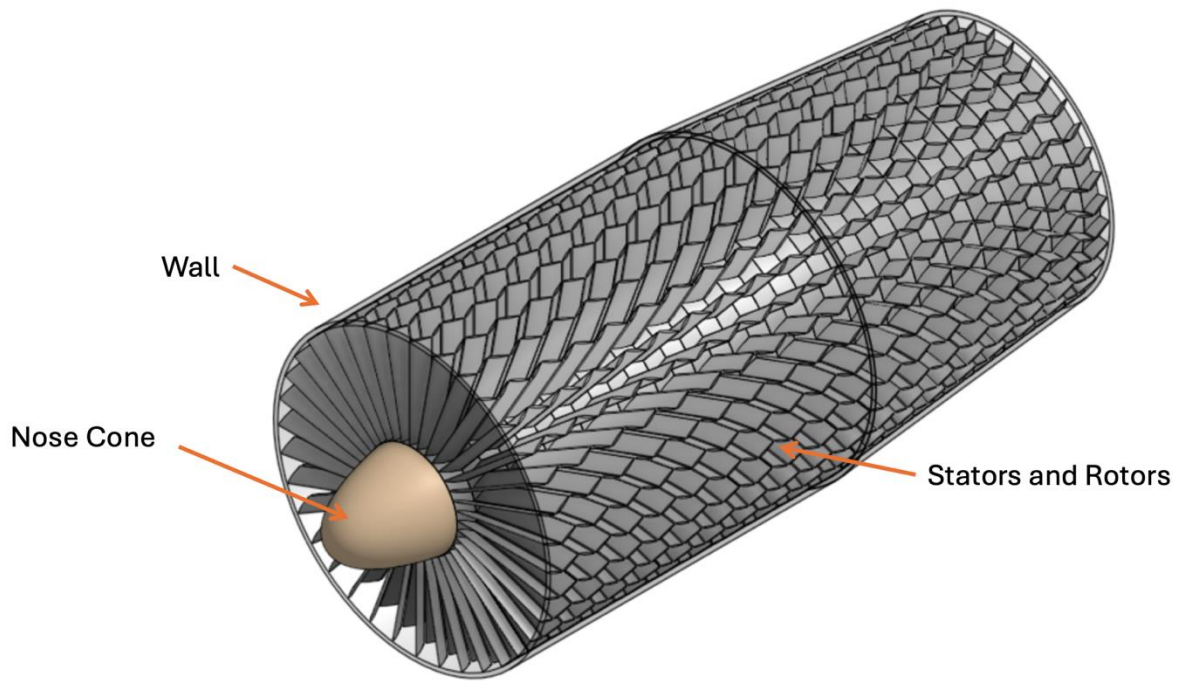


Figure 2. Overall Compressor Design- Isometric View

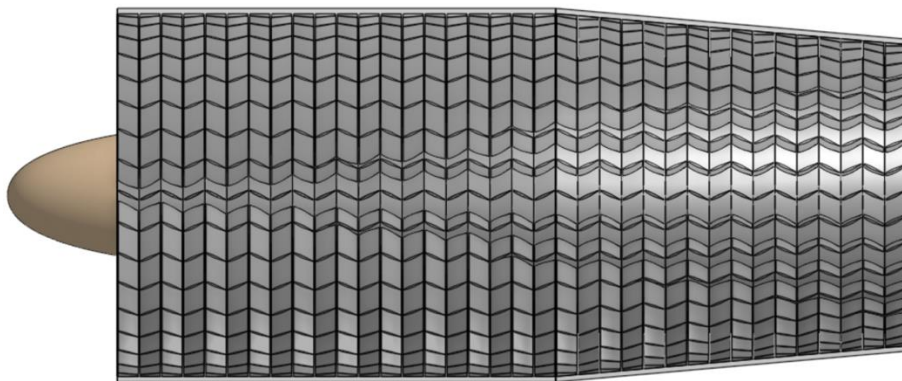


Figure 3. Overall Compressor Design – Side View

3. Performance Requirement

Compressor Design Specifications

Property	Value	Unit
Minimum pressure ratio	22:1	-
Maximum working altitude	40,000	ft
Minimum outer diameter	4	ft
Maximum outer diameter	6	ft
Maximum operating speed	40,000	rpm

Table 1. Compressor Design Specifications

4. Design Assumptions

4.1. Operating Conditions

Property	Value	Unit
Rotor speed	4,000	rpm

Table 2. Compressor Operating Conditions

4.2. Properties of Air at 40,000 ft

Property	Value	Unit
Temperature	-69.34	°F
Pressure	2.720	psi
Density	5.855e-4	lbm/ft ³
Speed of sound	1114.67	ft/s
Heat Capacity	0.240	Btu/lbm-R
Specific Heat Ratio	1.4	-

Table 3. Properties of Air at 40,000 ft

(Source: Aerospace, Mechanical & Mechatronic Engineering, University of Sydney)

4.3. Properties of Air at Sea Level

Property	Value	Unit
Temperature	59	°F
Pressure	14.696	psi
Density	2.378e-3	lbm/ft ³
Speed of sound	966.53	ft/s
Heat Capacity	0.240	Btu/lbm-R
Specific Heat Ratio	1.4	-

Table 4. Properties of Air at Sea Level

(Source: Aerospace, Mechanical & Mechatronic Engineering, University of Sydney)

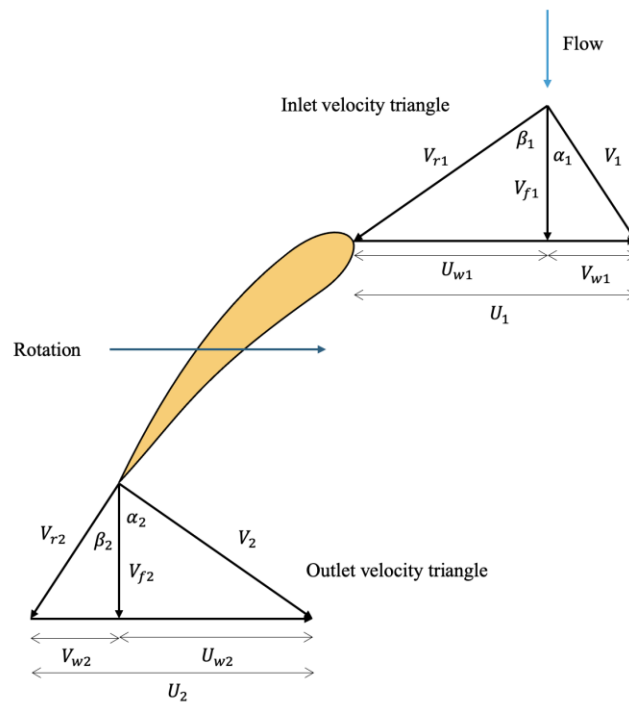
5. Hand Verification Calculation

To simplify hand verification calculation, a few assumptions were made:

- The compressor is operated in isentropic condition.
- The axial velocity of the air is constant throughout the compressor.
- Turbulence is negligible.
- The air acts as an ideal gas.

5.1. Temperature

In order to estimate the inlet and outlet temperature at each stage, some velocity components should be calculated using a velocity triangle, which is shown in Figure 4. Each velocity symbol and corresponding parameter is listed in Table 5.



This is a great representation of the input values and calculations!

Figure 4. Velocity Triangle for Blade

Symbol	Parameter
U	Linear velocity of rotor
U_w	Tangential velocity of rotor
V	Absolute velocity of air
V_f	Normal velocity of air
V_r	Relative velocity of air to blade
V_w	Swirl velocity of air
α	Fluid angle (alpha)
β	Fluid angle (beta)

Table 5. Symbols in Velocity Triangle

U_{w1} is determined by the angular velocity of the rotor and the length of the blade:

$$U_{w1} = (\omega_{rpm,rotor})(l_{blade})\left(\frac{2\pi}{60}\right)$$

With the known V_{f1} , V_{r1} can be calculated as following:

$$V_{r1} = \sqrt{U_{w1}^2 + V_{f1}^2}$$

The entry and exit angle of the fluid, β_1 and β_2 are calculated below, where $\Delta\beta$ is the change in angle of the blade (i.e., Angle of Attack), as well as U_{w2} and V_{r2} . Due to the assumption made above, $V_{f1} = V_{f2}$:

$$\beta_1 = \cos^{-1}\left(\frac{V_{f1}}{V_{r1}}\right), \text{ and } \beta_2 = \beta_1 + \Delta\beta$$

$$U_{w2} = V_{f2} \tan(\beta_2)$$

$$V_{r2} = \frac{V_{f2}}{\cos(\beta_2)}$$

V_{w2} can be calculated using the tangential velocity of the air at the entry and exit, which is a key component to find the absolute velocity of the air at the exit of the stage.

$$V_{w2} = U_{w1} - U_{w2}$$

$$V_2 = \sqrt{V_{f2}^2 + V_{w2}^2}$$

Calculating the temperature at each stage can be separated into two parts: from the entry to the first stage and the stage between stage.

From Stage 0 (entrance) to Stage 1, the temperature at Stage 1 can be easily calculated using the following equation, where V_0 is the initial inlet velocity of the air ($=V_f$) and $c_{p,air}$ is the heat capacity of the air.

$$T_1 = T_0 + \frac{V_0^2}{c_{p,air}}$$

From Stage 1 to Stage 2 (and onward),

$$\tau_{rotor} = \dot{m}_{air} * l_{blade} * |V_{w2}|$$

$$P_{rotor} = \tau_{rotor} * \omega_{rotor}$$

$$w_{rotor} = \frac{P_{rotor}}{\dot{m}_{air}}$$

$$T_2 = T_1 + \frac{w_{rotor}}{c_{p,air}}$$

where τ_{rotor} = work done by the single blade of a rotor at each stage,

$\dot{m}_{air}$ = mass flow rate of the air,

Add in sample calculations for values. For the hand calculations use the ideal gas law and formulate for there?

P_{rotor} = shaft power,

and w_{rotor} = specific work of the single blade of a rotor.

5.2. Pressure & Compressor Pressure Ratio (CPR)

The compressor pressure ratio (CPR) is calculated using the isentropic relations. The exit pressure is simply the multiplication of the entrance pressure and the CPR.

$$\frac{P_2}{P_1} = CPR_{1,2} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$P_2 = (P_1)(CPR_{1,2})$$

5.3. Number of Stages

The minimum number of stages needed to meet the design requirement is the product of the compressor pressure ratio at each stage.

$$CPR_{comp} = \prod CPR_{i,i+1}$$

Based on the calculation, it was concluded that to meet the CPR design requirement, 16 stages are needed when operating at 40,000 ft. However, when the compressor is operated at Sea Level with the current design, the compressor doesn't meet the minimum pressure ratio with more than 20 stages. If one wants to operate the compressor at Sea Level, it would be recommended to modify the compressor design. Below are the tabulated results of the estimated entrance and exit temperature and pressure of the air, total compressor pressure ratio, and the number of the stages. Full data can be found in Appendix A.

Stage (at 40,000 ft)	Entrance temperature [°F]	Exit temperature [°F]	Entrance pressure [psi]	Exit pressure [psi]	CPR
1	-69.34	34.11	2.72	6.19	2.277
2	34.11	109.60	6.19	10.19	1.645
⋮	⋮	⋮	⋮	⋮	⋮
15	456.04	477.67	53.80	58.38	1.085
16	477.67	496.95	58.38	62.69	1.074
Total CPR					23.044

Table 6. Tabulated Result of Temperature, Pressure, and CPR at 40,000 ft

Watch significant
figures

Stage (at Sea Level)	Entrance temperature [°F]	Exit temperature [°F]	Entrance pressure [psi]	Exit pressure [psi]	CPR
1	59.00	162.45	14.70	21.14	1.439
2	162.45	187.01	21.14	24.21	1.145
⋮	⋮	⋮	⋮	⋮	⋮
19	619.14	630.18	145.18	150.45	1.036
20	630.18	638.08	150.45	154.30	1.026
Total CPR					10.500

Table 7. Tabulated Result of Temperature, Pressure, and CPR at Sea Level

5.4. Compressor Efficiency

The efficiency of the compressor is the ratio between the total power due to the pressure increase throughout the compressor and the total mechanical power by the movement of the rotor. The total power due to the pressure increase can be calculated by summing the power due to the pressure increase at each stage, and the total mechanical power by the rotor's movement can be calculated by summing the power by the rotor's movement at each stage:

$$\eta = \frac{\dot{W}_{pressure,total}}{\dot{W}_{rotor,total}} = \sum \frac{\dot{W}_{pressure,stage}}{\dot{W}_{rotor,stage}}$$

The power due to the pressure increase at each stage can be calculated using the following formula:

$$\dot{W}_{pressure,stage} = \frac{\dot{m}_{air} \Delta P}{\rho_{air}}$$

where $\dot{m}_{air}$ is the mass flow rate of the air at the entrance, ΔP is the pressure difference at each stage, and ρ_{air} is the density of the air.

The power by the movement of the rotor can be calculated as following:

$$\dot{W}_{rotor,stage} = \tau \omega = F_D l_{blade} (\# \text{ of blade}) \omega$$

where $F_D = \frac{1}{2} C_D \rho_{air} U^2 A$ with $U = V_{f1}$ and A = cross-sectional area of the blade. Considering the blade a flat plate, $C_D = 1.28 \sin |90^\circ - \beta_1|$.

Based on the calculation, it was determined that the total efficiency of the compressor at 40,000 ft and the sea level is 59.20% and 48.92%, respectively.

Power calculated at both sea level and 40,000 ft?

6. Conceptual Design

6.1. Blade, Rotor, and Stator

As shown in Figure 5, implementing the NACA 65-210 airfoil profile, the geometry with the chord length of 4-inch was selected as a cross-sectional area of the blade. Then the geometry was extruded up to the desired length for each stage. The blade length at each stage can be found in Appendix B.

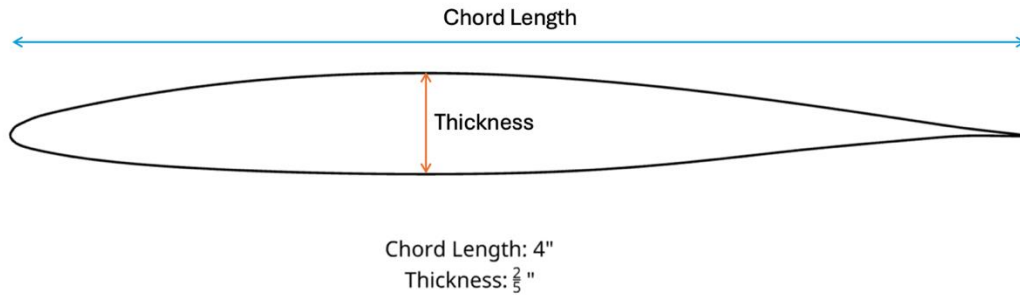


Figure 5. Cross-section View – Blade

With the twist angle of 25° , 36 blades were attached to the rotor and stator. The thickness of the rotor and stator is 4 inches. Figures 6 and 7 are the front view with dimensions and isometric view of the rotor at Stage 1, respectively.

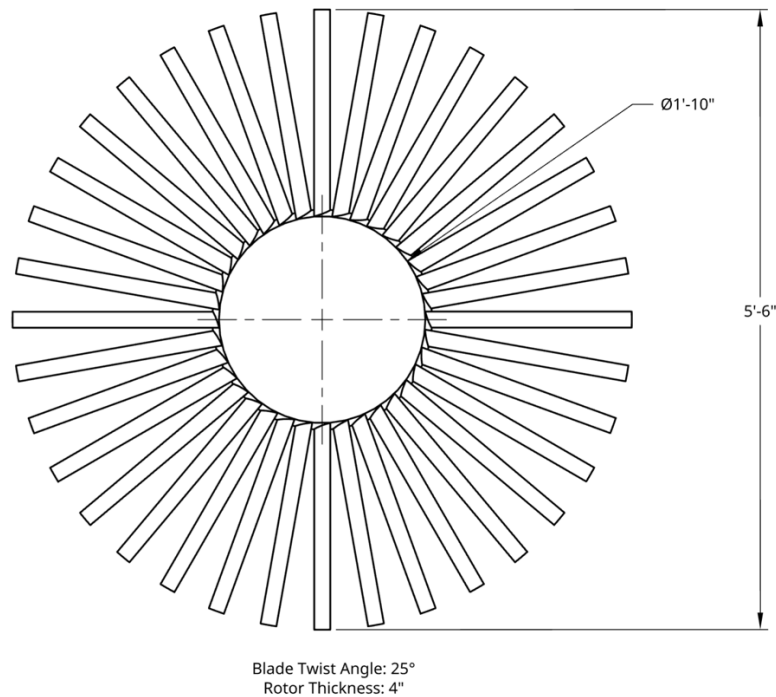


Figure 6. Drawing – Rotor at Stage 1

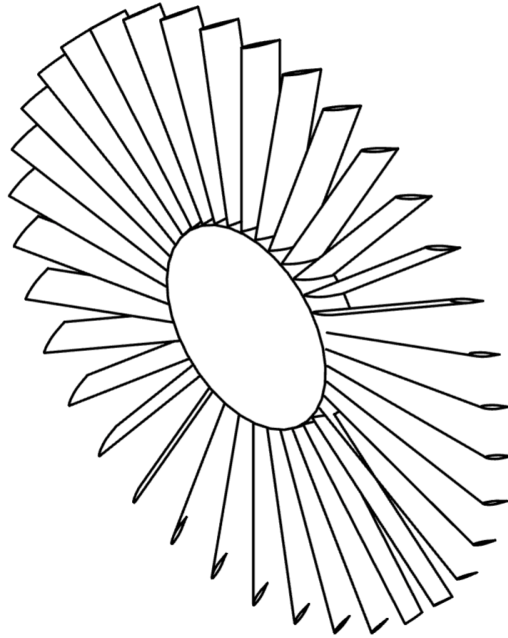


Figure 7. Isometric View – Rotor at Stage 1

6.2. Nose Cone

The nose cone is attached to the rotor at Stage 1. Figures 8 and 9 are the drawing and isometric view of the nose cone, respectively.

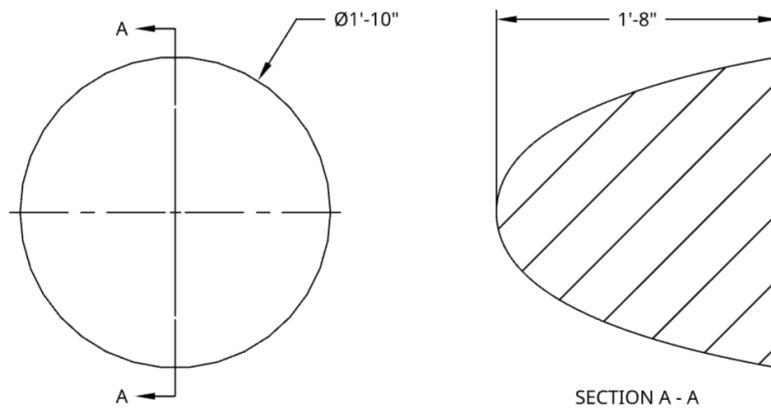


Figure 8. Drawing – Nose Cone

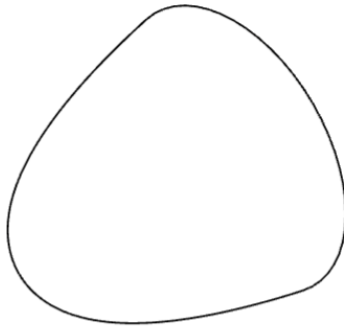


Figure 9. Isometric View – Nose Cone

6.3. Wall

The wall, with the 2-inch thickness, was designed to protect the rotors and stator from any external damage. The total length of the wall depends on the number of the stages. Figure 10 is the drawing of the compressor wall, and Figure 11 shows the isometric view of the wall.

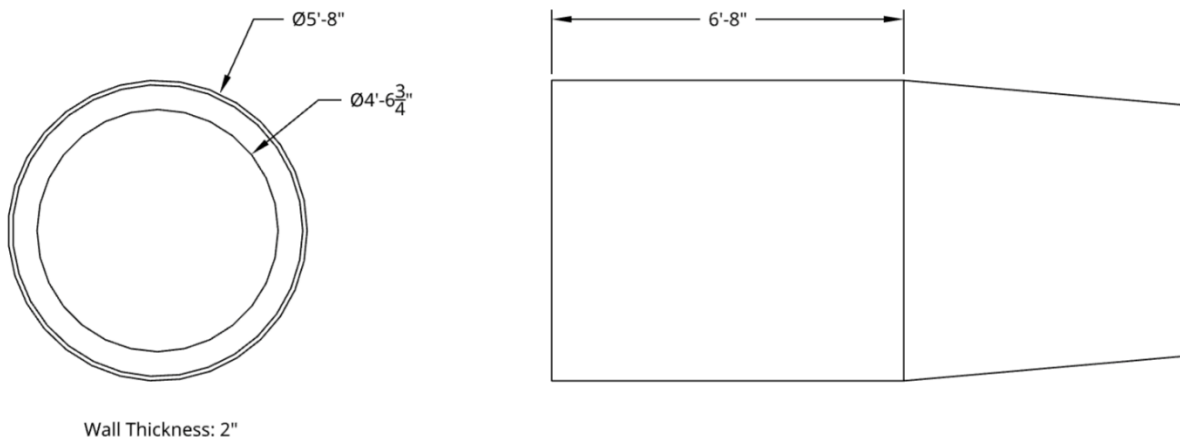


Figure 10. Drawing – Wall

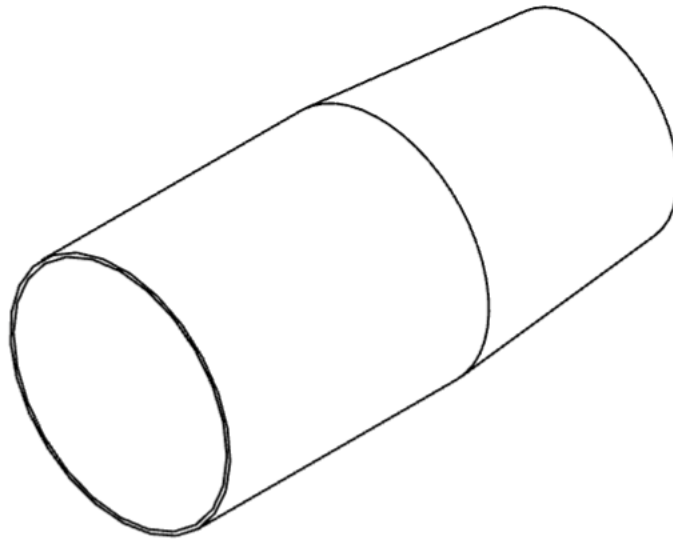


Figure 11. Isometric View – Wall

6.4. Overall Size

Property	Value
Inlet diameter	5 ft 8 in
Outlet diameter	4 ft 6 $\frac{3}{4}$ in
Length	12 ft $\frac{2}{3}$ in
Number of stages	19

Table 8. Overall Size of Compressor

Notes: Length depends on the number of the stages attached.

7. CFD Simulations

7.1. Mesh

HyperMesh 2022 was utilized for meshing. To optimize the CFD simulation, a single stator and rotor section was meshed at a time. To ensure accurate results, mesh refinement was applied near the leading and trailing edge of the two blades. Table 9 shows the parameters of the rotor-stator section mesh, and Figure 12 shows the final mesh of the first stage simulation. The subsequent stages simulated were meshed using a similar approach. The first stage mesh was made up of 376,487 elements. The aspect ratio of the mesh was above 5 for all elements.

Property	Value
Method	Tetrahedral
Minimum Element Size	0.075 in
Average Element Size	0.125 in

Table 9. Mesh Setup Stage 1

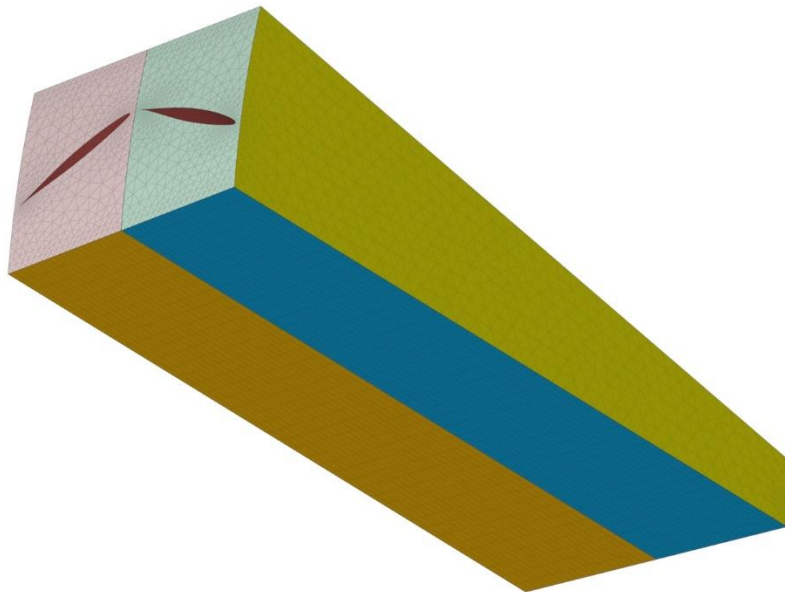


Figure 12. Rotor-Stator at Stage 1 Mesh

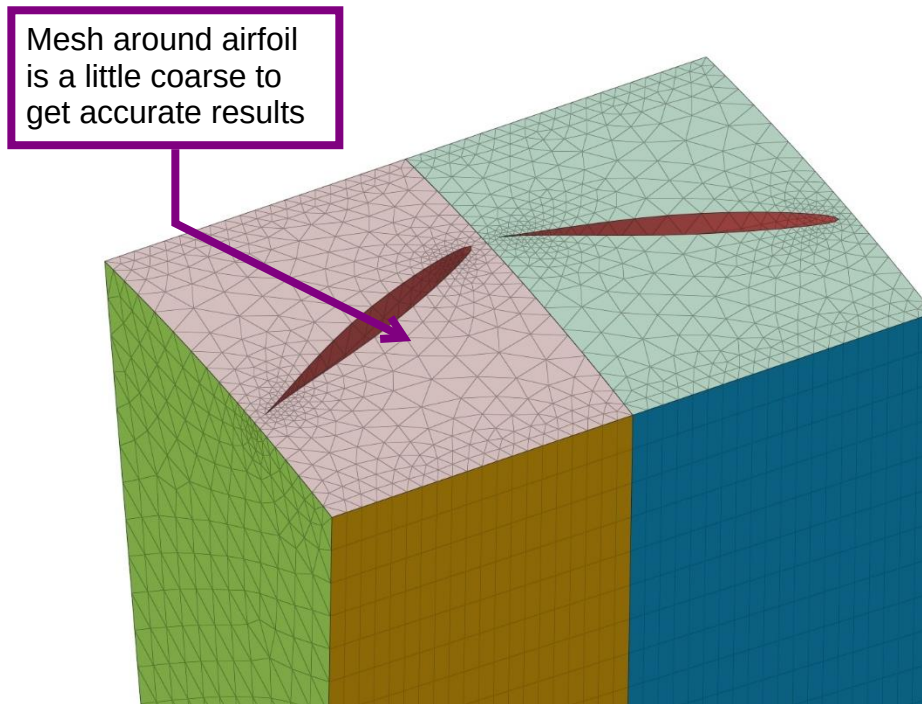


Figure 13: Zoomed-In Rotor-Stator Mesh



Figure 14: Top View Section Cut of Rotor-Stator Mesh

Also, it is notable that the later stages are of a smaller size. This means that they were meshed finer to produce more accurate results because element count could be kept around the same by making elements smaller. Below is a view of stage 11 which is when the compressor shroud begins to decrease in diameter while keeping the hub at the same diameter.

Property	Value
Method	Tetrahedral
Minimum Element Size	0.05 in
Average Element Size	0.125 in

Table 10. Mesh Setup at Stage 11

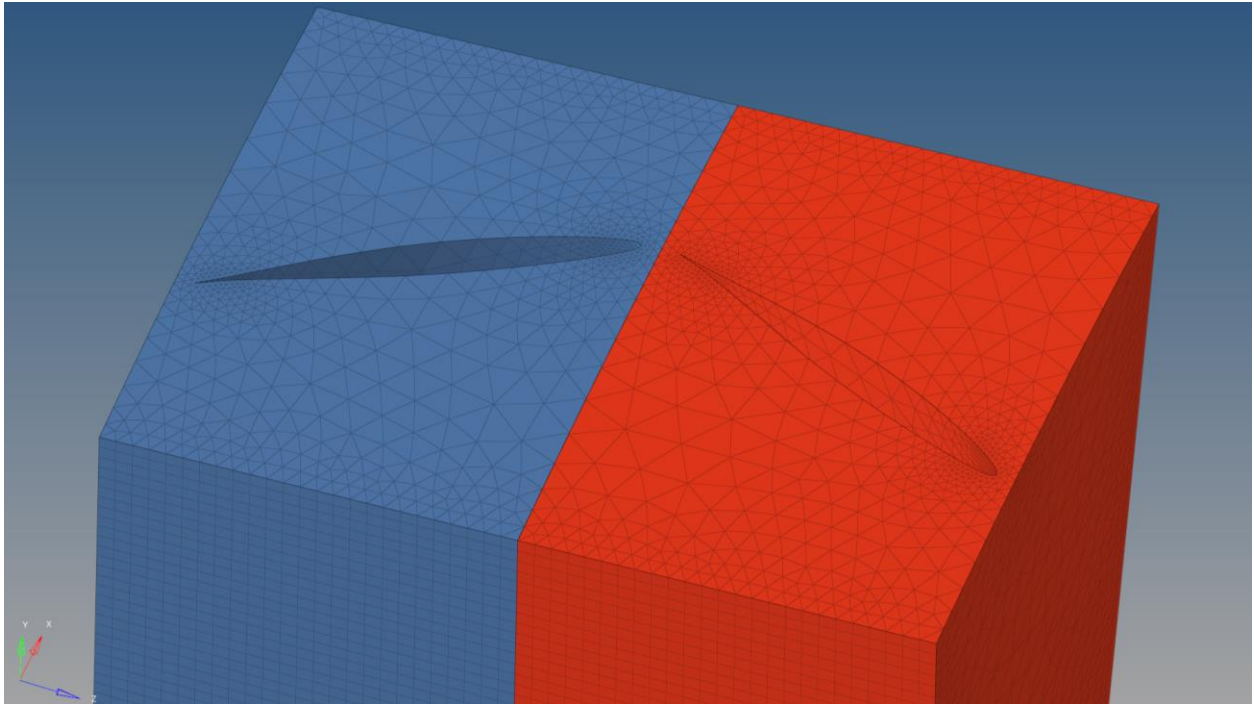


Figure 15: Orthogonal View of Rotor-Stator Mesh at Stage 11

The mesh looks very coarse. Use edge sizing and inflation around the full airfoil. Use a curvilinear (body-fitted) mesh around the air foil. For the interior of the rotor and stator, use a polyhedral or hexahedral mesh instead of a tetrahedral mesh.

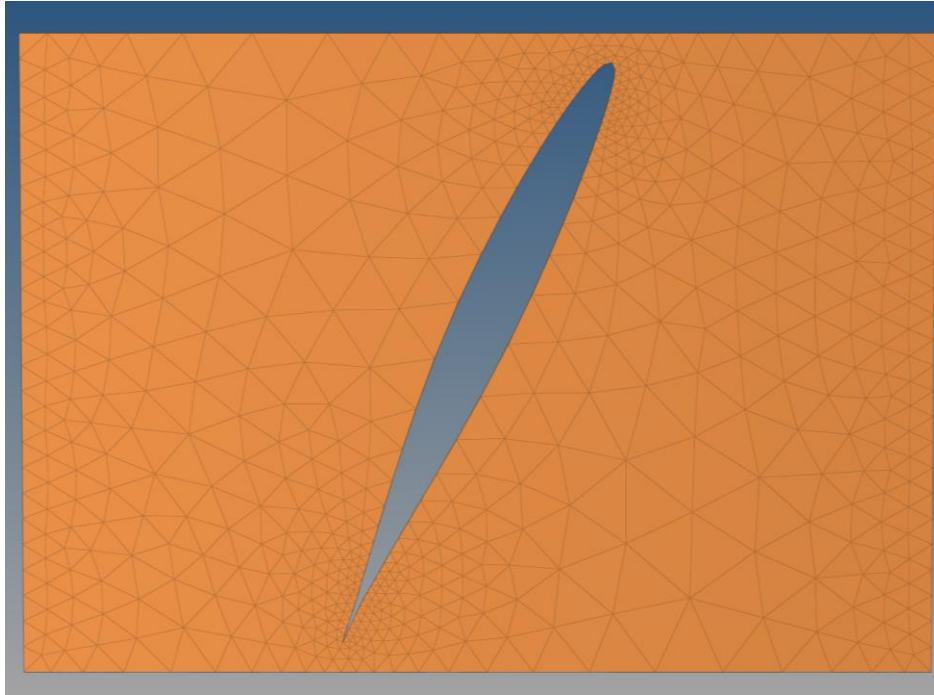


Figure 16: Top View of Rotor Mesh at Stage 11

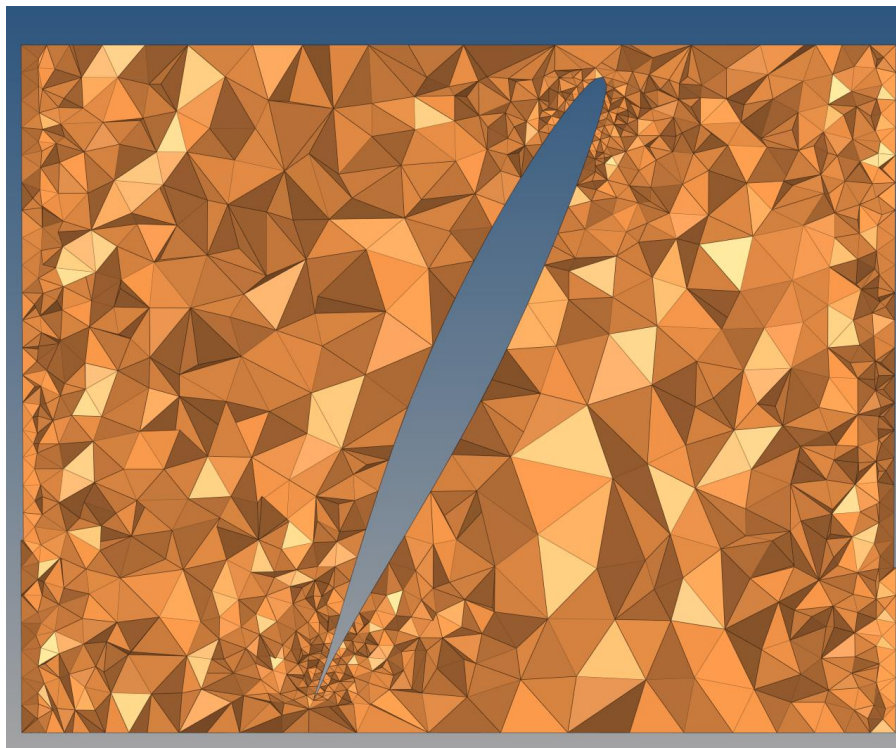


Figure 17: Top View Section Cut of Rotor Mesh at Stage 11

7.2. Boundary Conditions

As shown in Figure 18. the boundary conditions consist of rotor and stator shrouds, rotor and stator blades, rotor inlet, and rotor and stator periodic conditions. Table 11 shows the example the boundary conditions in Ansys Fluent, Stage 1 at Sea Level.

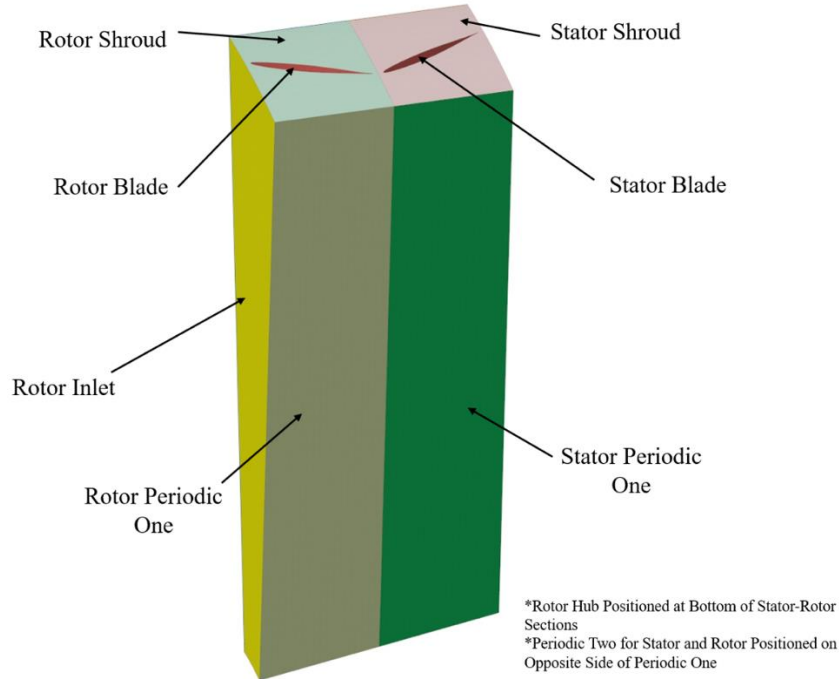


Figure 18. Boundary Conditions

Boundary Condition	Tab	Property	Value
Cell Zone	-	Fluid	Air*
Inlet (Rotor)	Momentum	Reference Frame	Absolute
		Gauge Total Pressure	14.7 psi
		Initial Gauge Pressure	14.5 psi
		Turbulent Intensity	1 %
		Hydraulic Diameter	21.69 in
	Thermal	Temperature	59 °F
Outlet (Stator)	Momentum	Backflow Reference Frame	Absolute
		Gauge Pressure	14.7 psi
		Radial Equilibrium Pressure Distribution	On
		Turbulent Intensity	1 %
		Hydraulic Diameter	21.69 in
	Thermal	Backflow Total Temperature	79.80 °F

Table 11. Tabulated Boundary Conditions – Stage 1 at Sea Level

* Air density was applied based on “Ideal Gas” option in ANSYS Fluent.

7.3. Solution Setup

The simulation was run using the Ansys Fluent with the parameters in Table 12. the number of the iterations was set to 1500.

Property	Tab	Setup
Model	Energy	On
	Viscous Model	K-epsilon – Standard – Enhanced Wall Treatment
Solution Method	Scheme	Pressure–Velocity Coupling – Coupled
	Gradient	Least Squares Cell Based
	Pressure	Second Order
	Momentum	Second Order Upwind
	Turbulent Kinetic Energy	Second Order Upwind
	Turbulent Dissipation Rate	Second Order Upwind
	Energy	Second Order Upwind
Time Advancement	Number of Time Steps	2,000 – 6,000
	Time Step Size	2.78e-5 – 2e-6 (s)
	Max Iterations/Time Step	15

Table 12. Solution Setup

Show this works with the mesh size

8. Simulation Results

8.1. Residuals

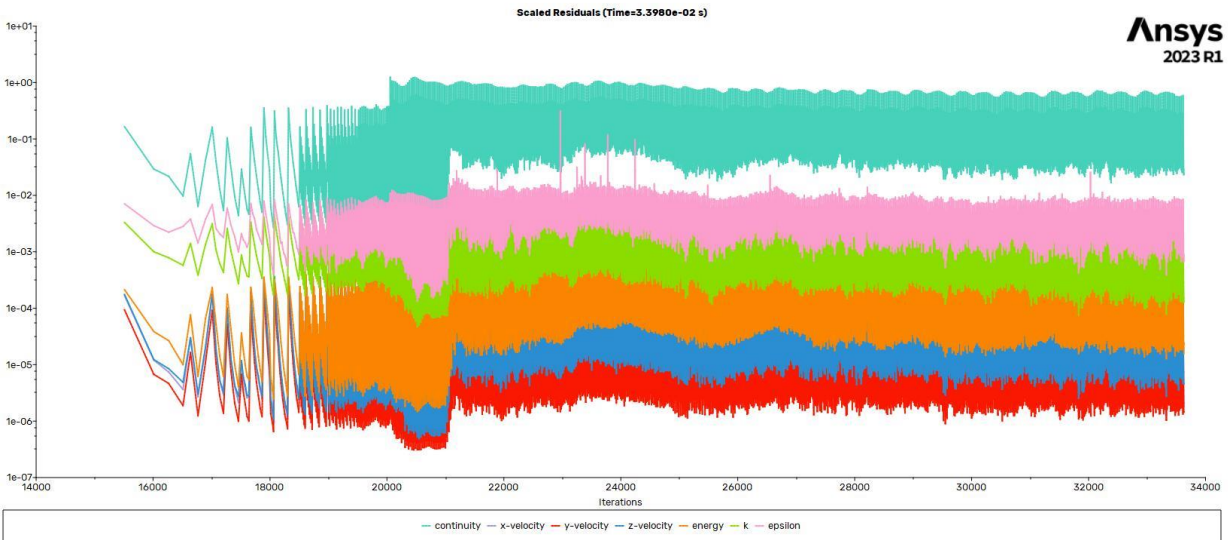


Figure 18. Residual vs Iterations Plot (Stage 1 at 40,000ft)

Figure 19. Residual vs Iterations Plot (Stage 2 at 40,000ft)

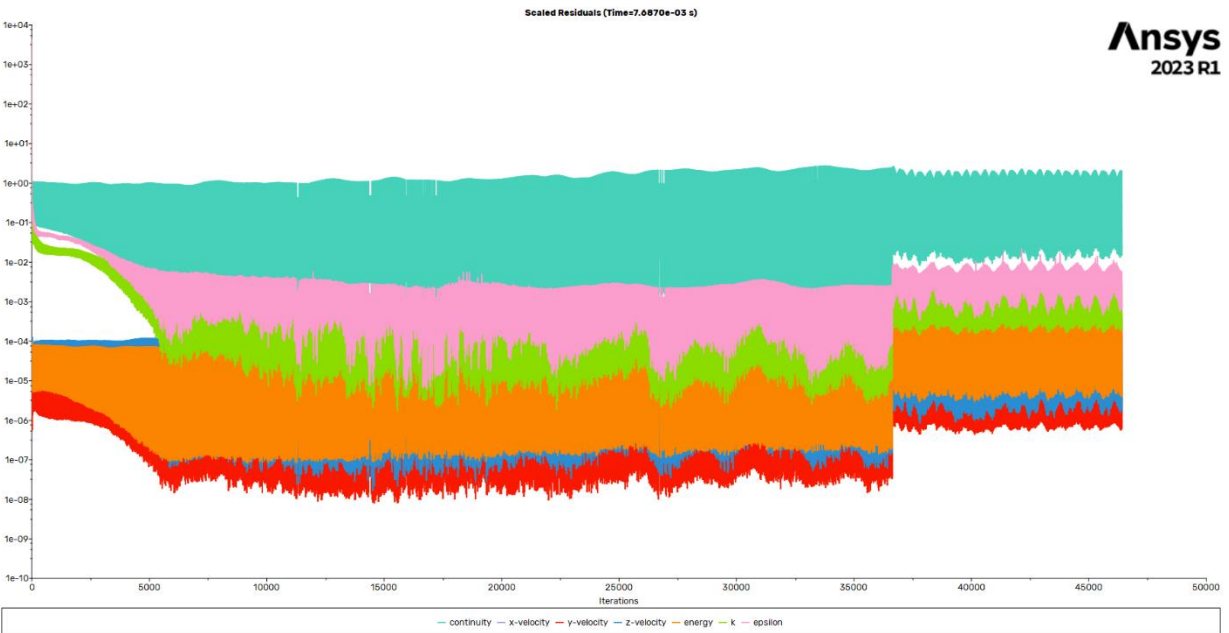
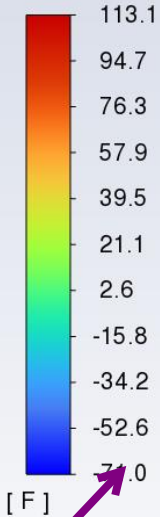


Figure 20. Residual vs Iterations Plot (Stage 11 at 40,000ft)

8.2. Temperature Contour Plots

contour-2
Static Temperature



[F]

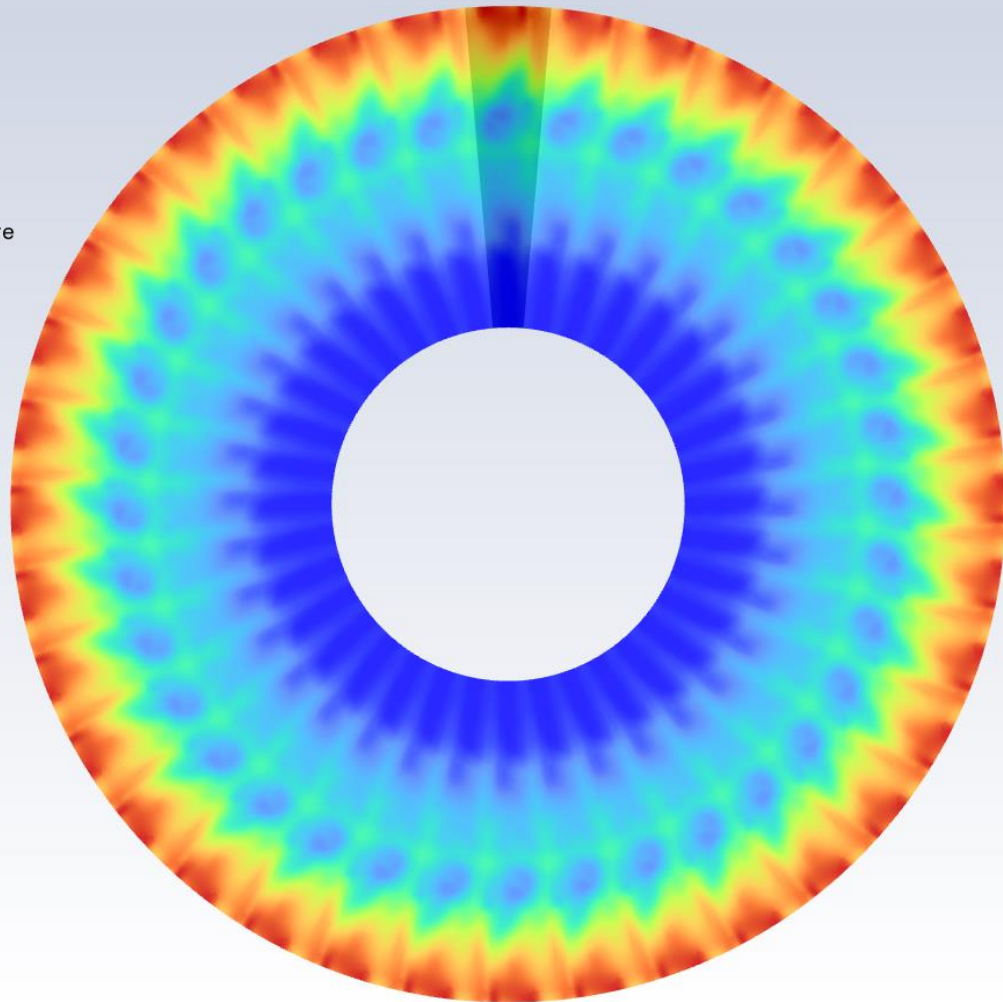


Figure 21. Temperature Contour of Rotor and Stator: Mid-section Plane (Stage 1 at 40,000 ft)

Are you sure about
-71F

You also need an isometric image
for this to be clear

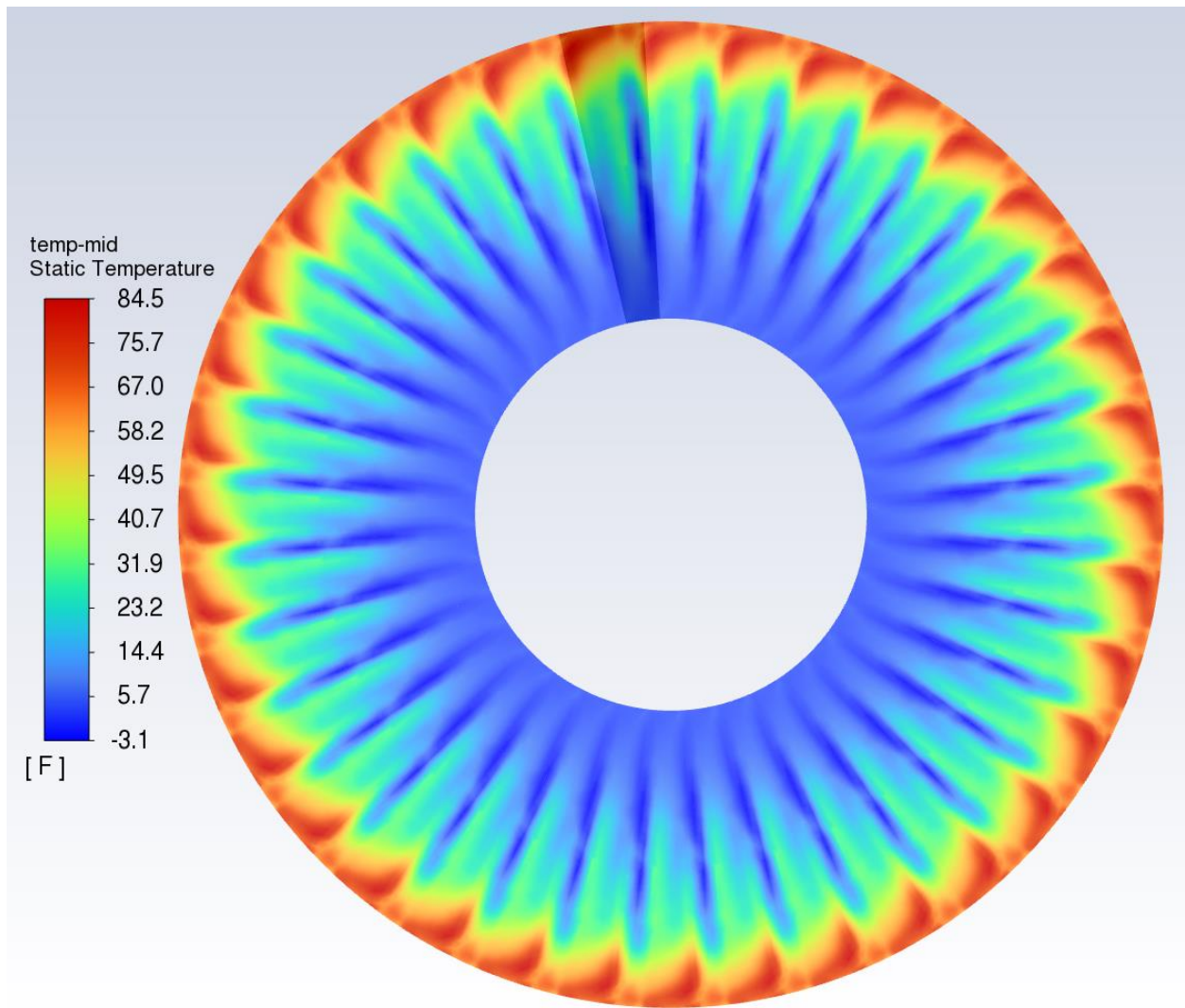


Figure 22. Temperature Contour of Rotor and Stator: Mid-section Plane (Stage 2 at 40,000 ft)

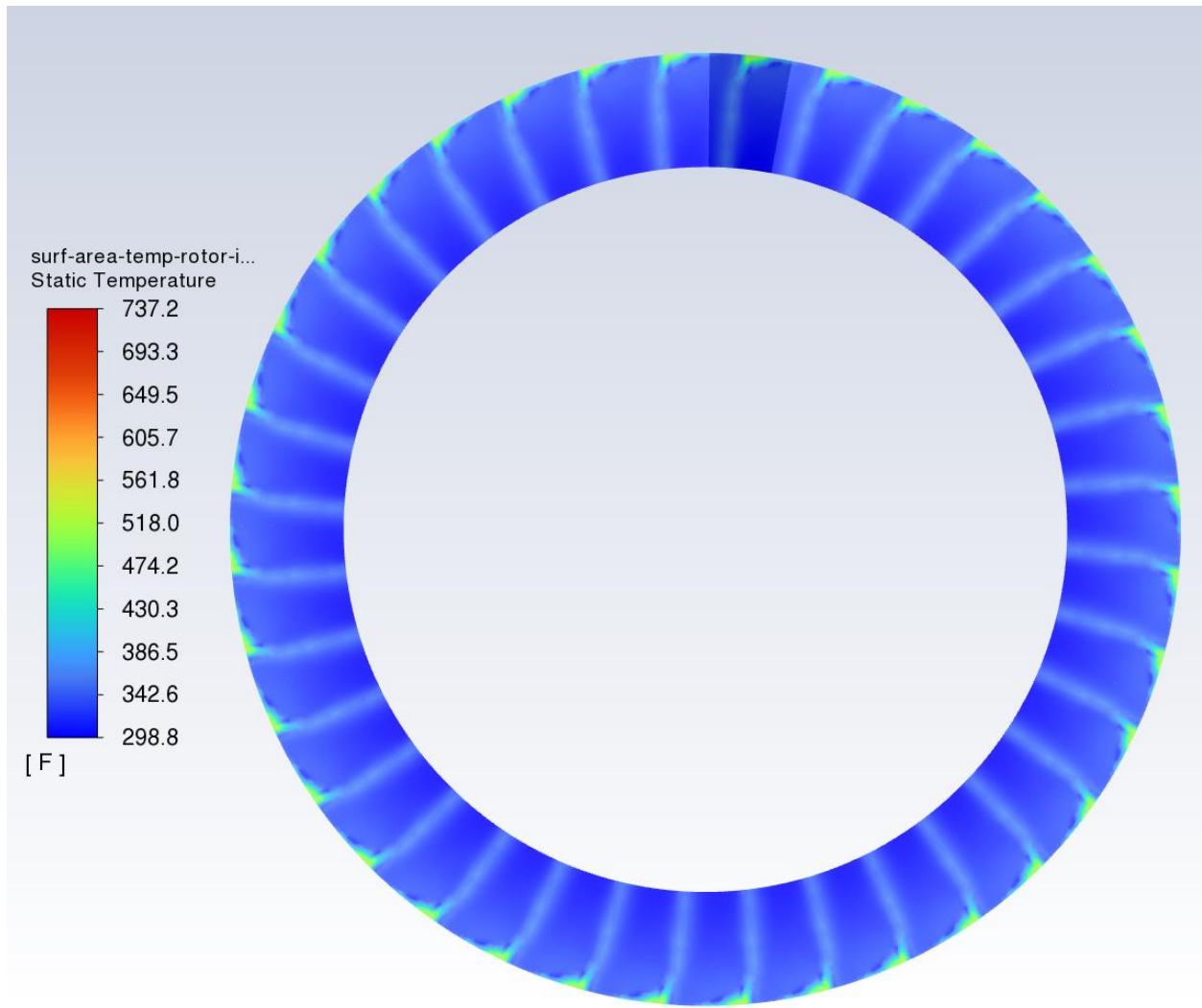


Figure 23. Temperature Contour of Rotor and Stator: Mid-section Plane (Stage 11 at 40,000 ft)

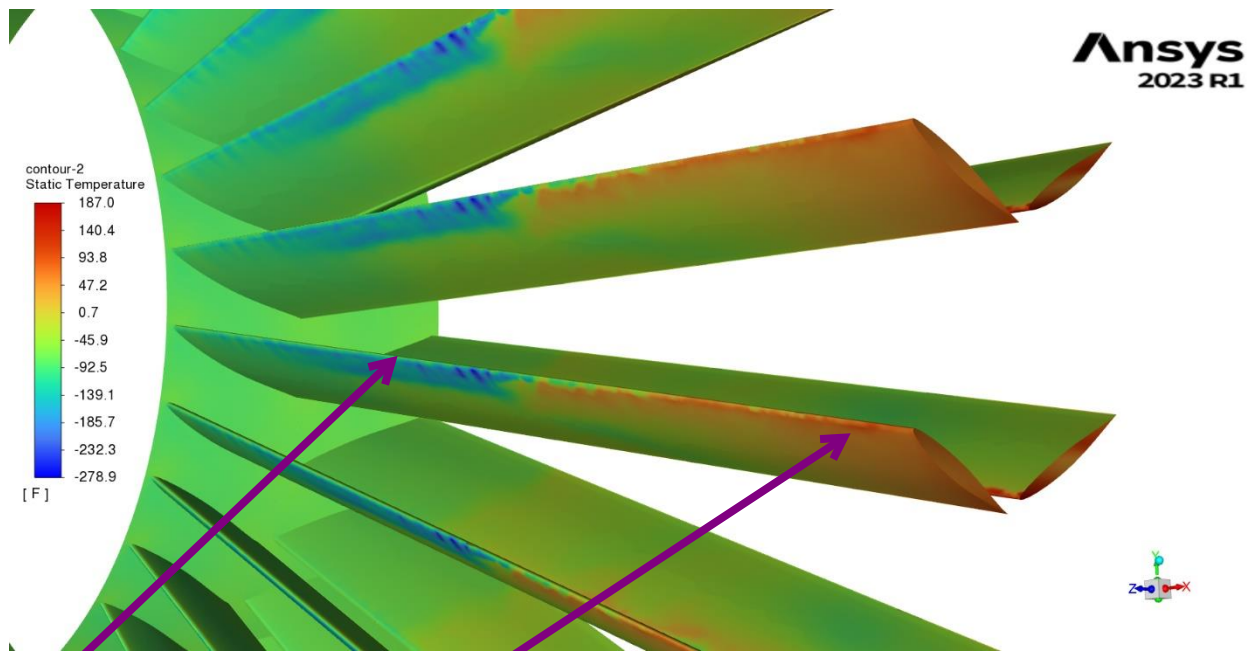


Figure 24. Temperature Contour of Rotor and Stator: Blades and Hub Walls (Stage 1 at 40,000 ft)

This looks
unphysical

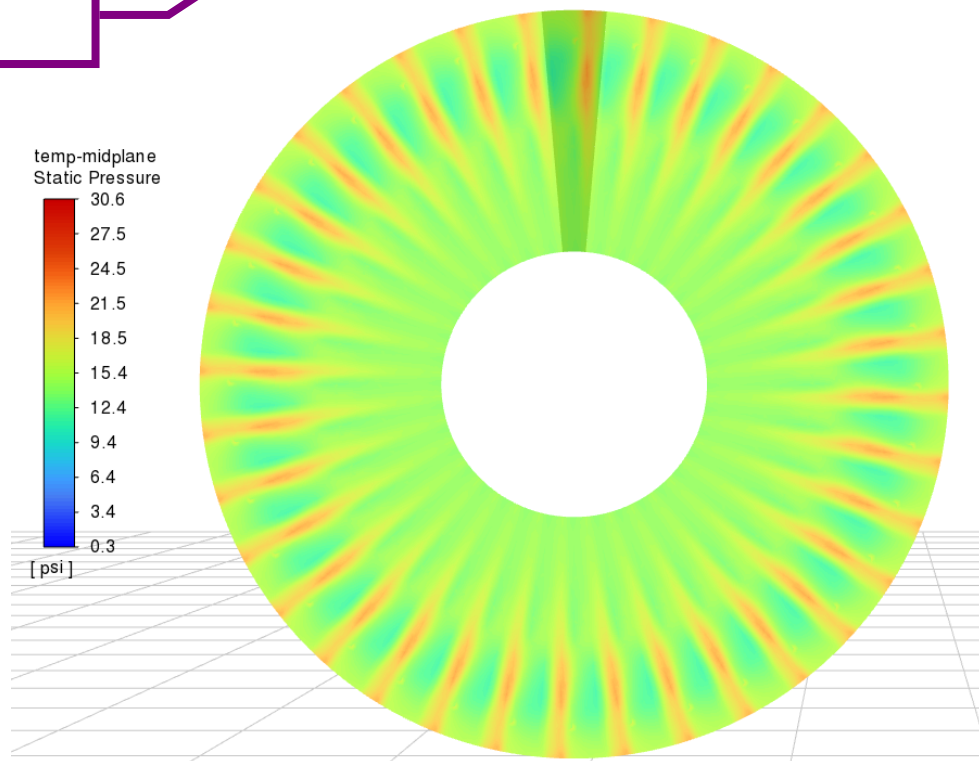


Figure 25. Temperature Contour of Rotor and Stator: Mid-section Plane (Stage 1 at Sea Level)

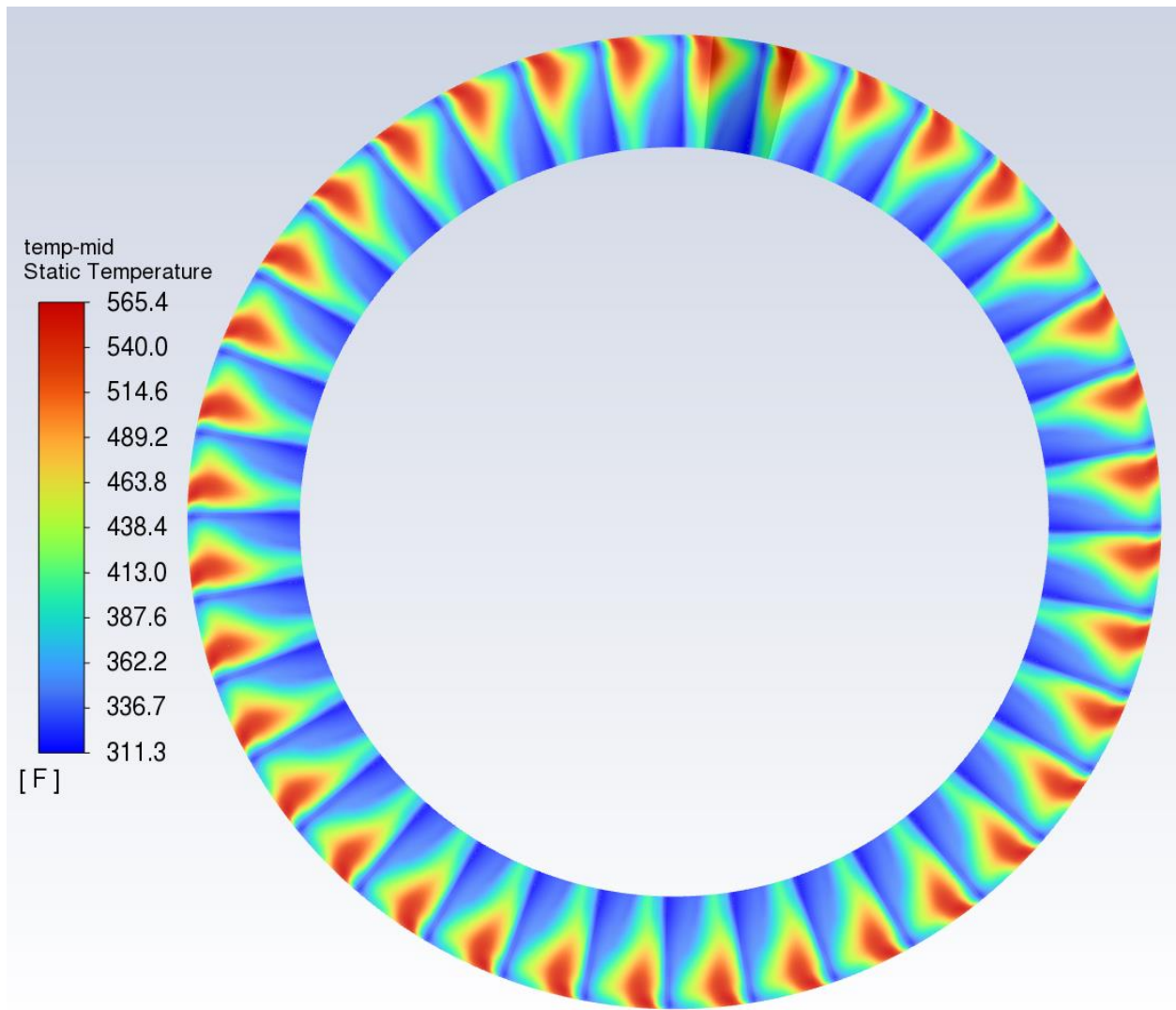


Figure 26. Temperature Contour of Rotor and Stator: Mid-section Plane (Stage 11 at Sea Level)

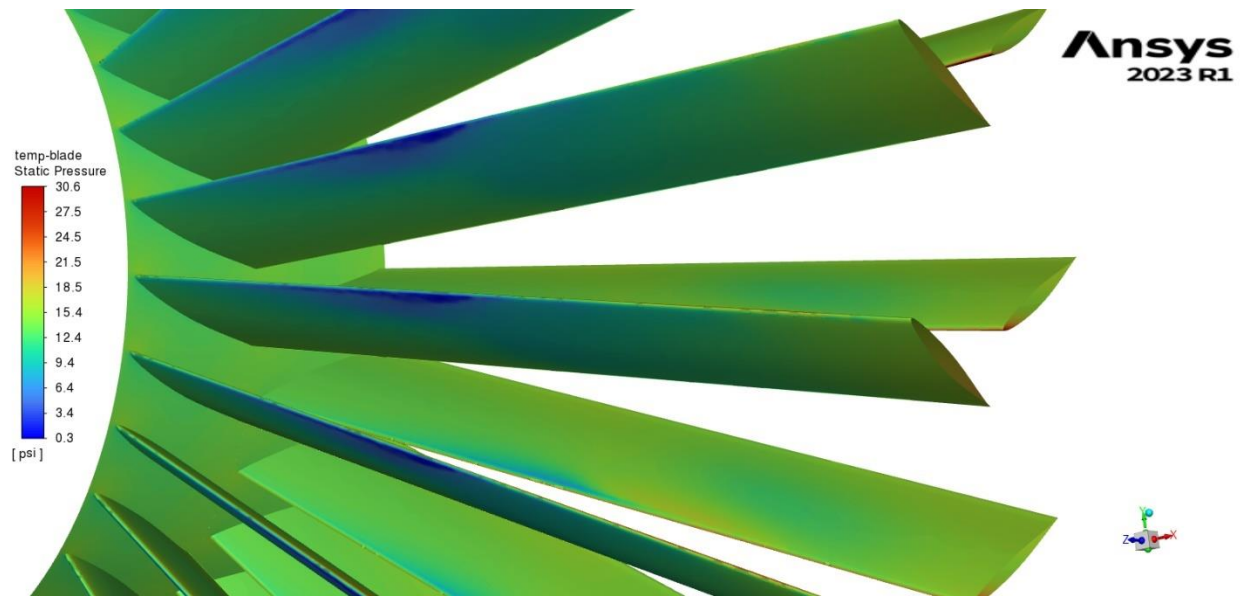


Figure 27. Temperature Contour of Rotor and Stator: Blades and Hub Walls (Stage 1 at Sea Level)

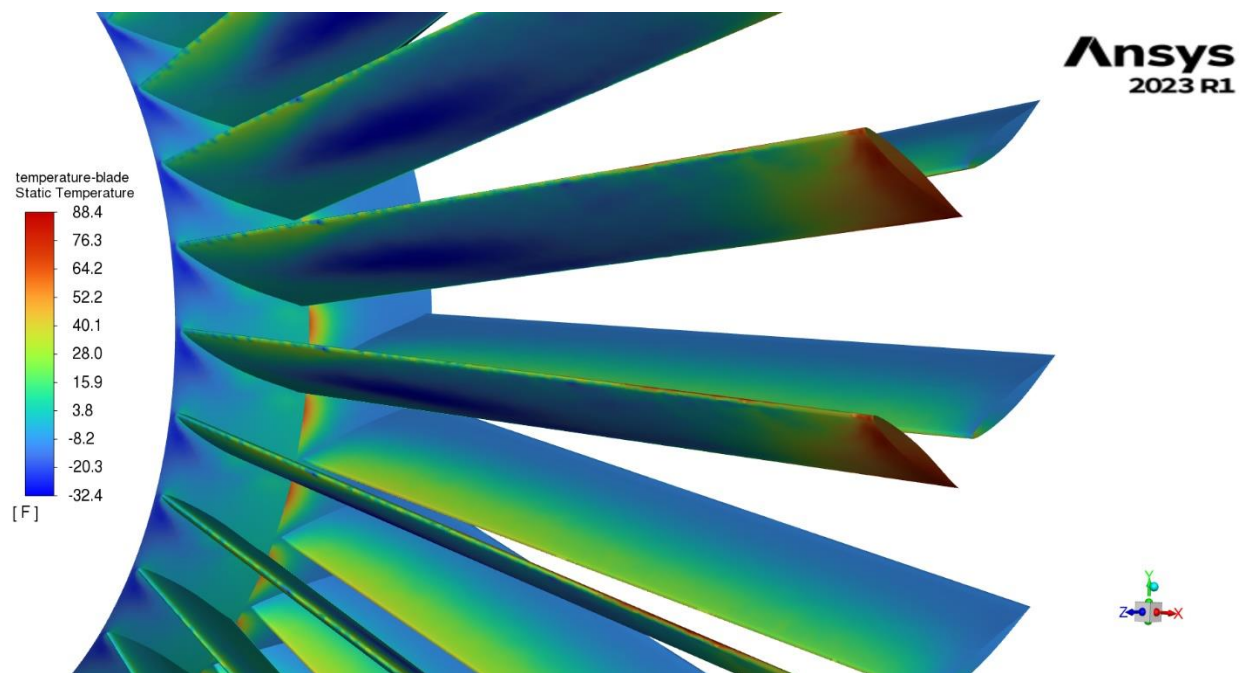


Figure 28. Temperature Contour of Rotor and Stator: Blades and Hub Walls (Stage 2 at 40,000 ft)

Temperature at
sea level is
more
reasonable and
realistic.

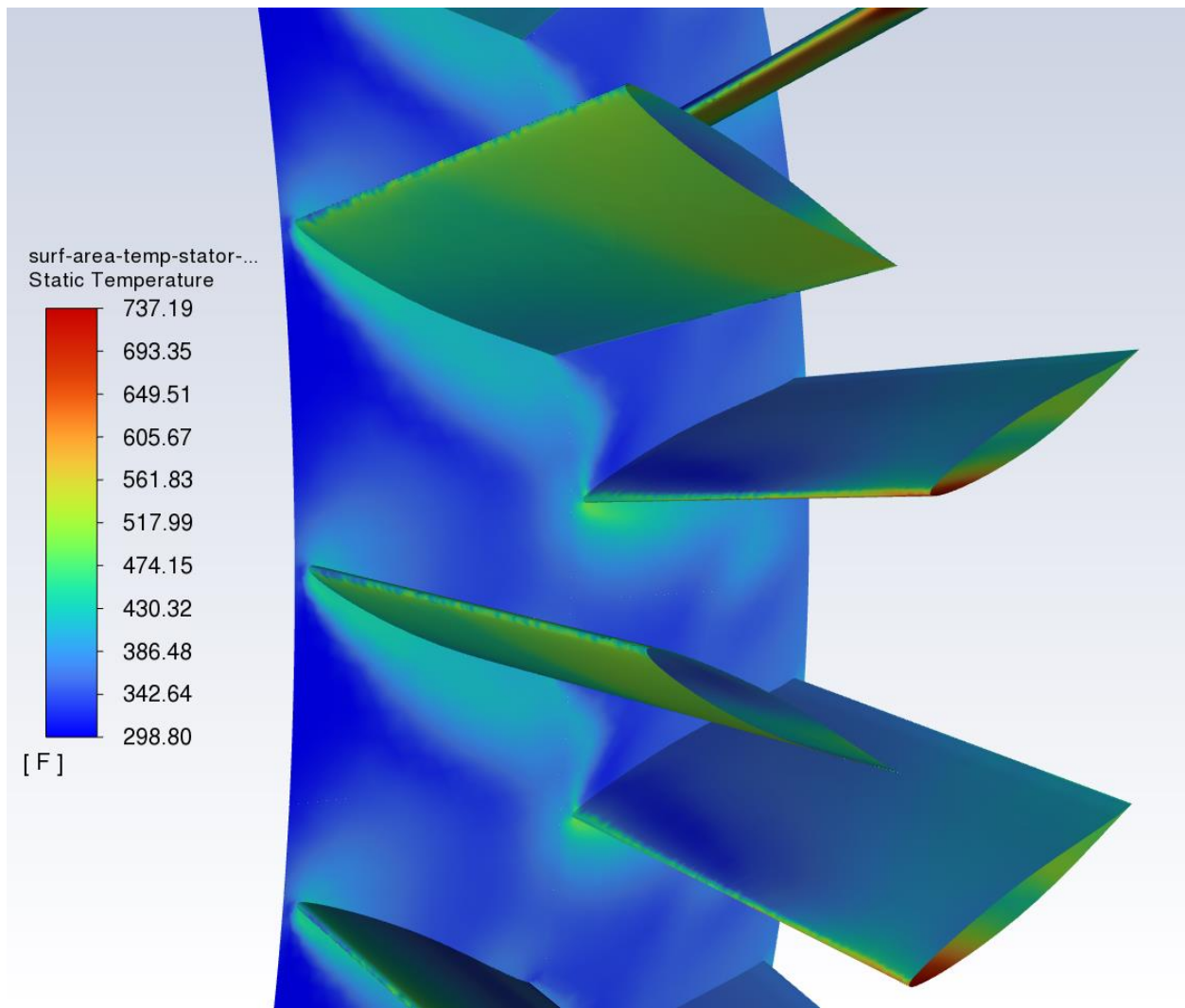


Figure 29. Temperature Contour of Rotor and Stator: Blades and Hub Walls (Stage 11 at 40,000 ft)

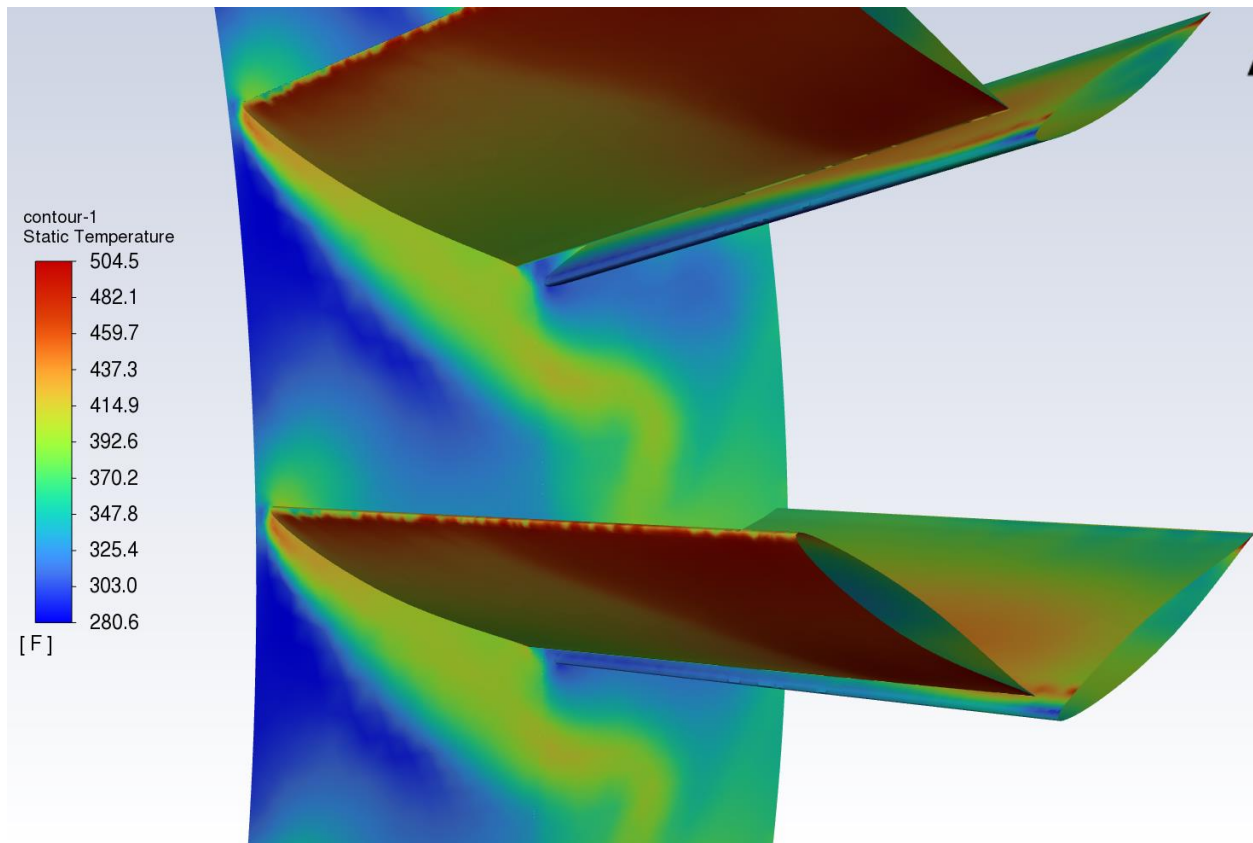


Figure 30. Temperature Contour of Rotor and Stator: Blades and Hub Walls (Stage 11 at Sea Level)

8.3. Pressure Contour Plots

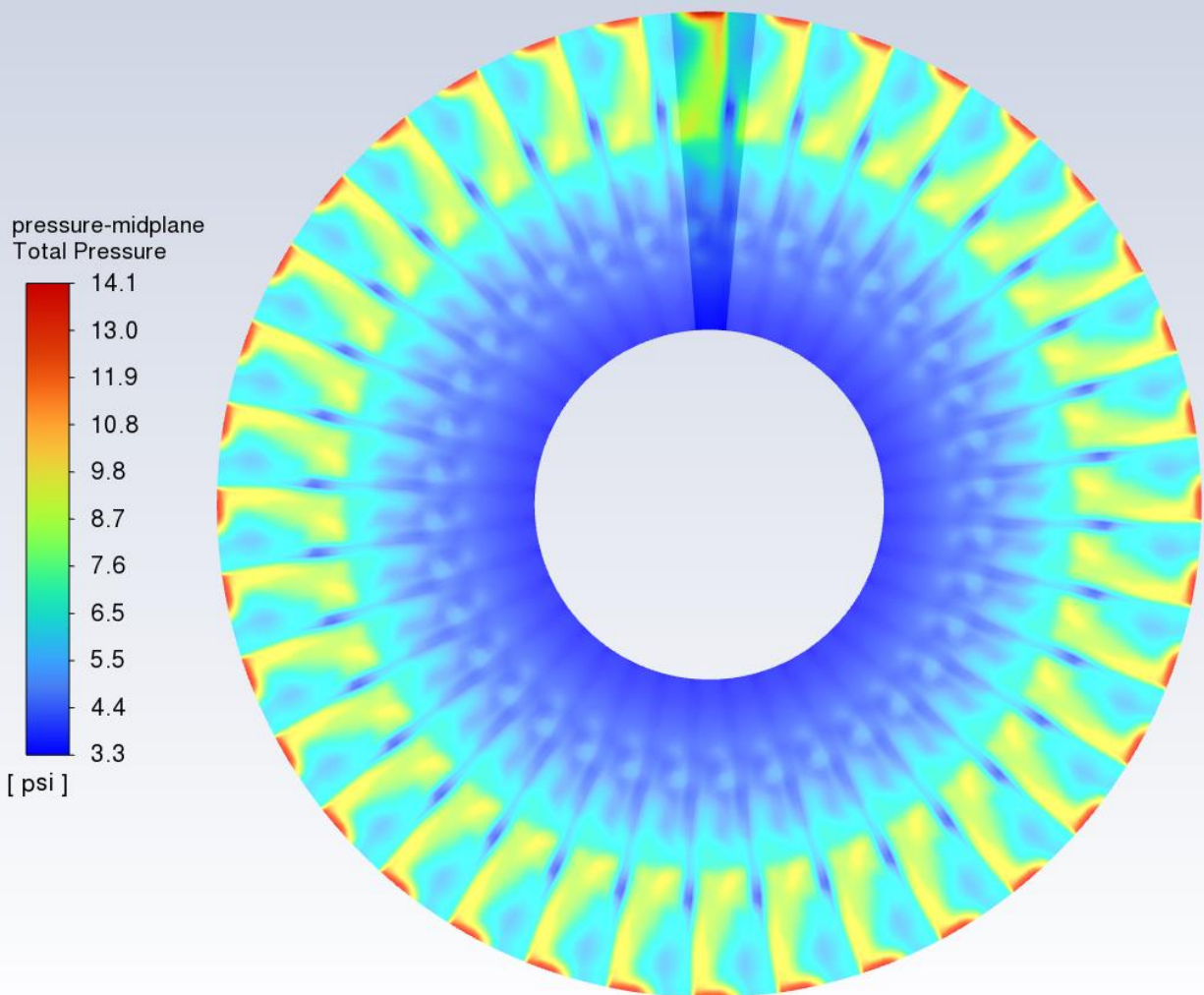


Figure 31. Pressure Contour of Rotor and Stator: Mid-section Plane (Stage 1 at 40,000 ft)

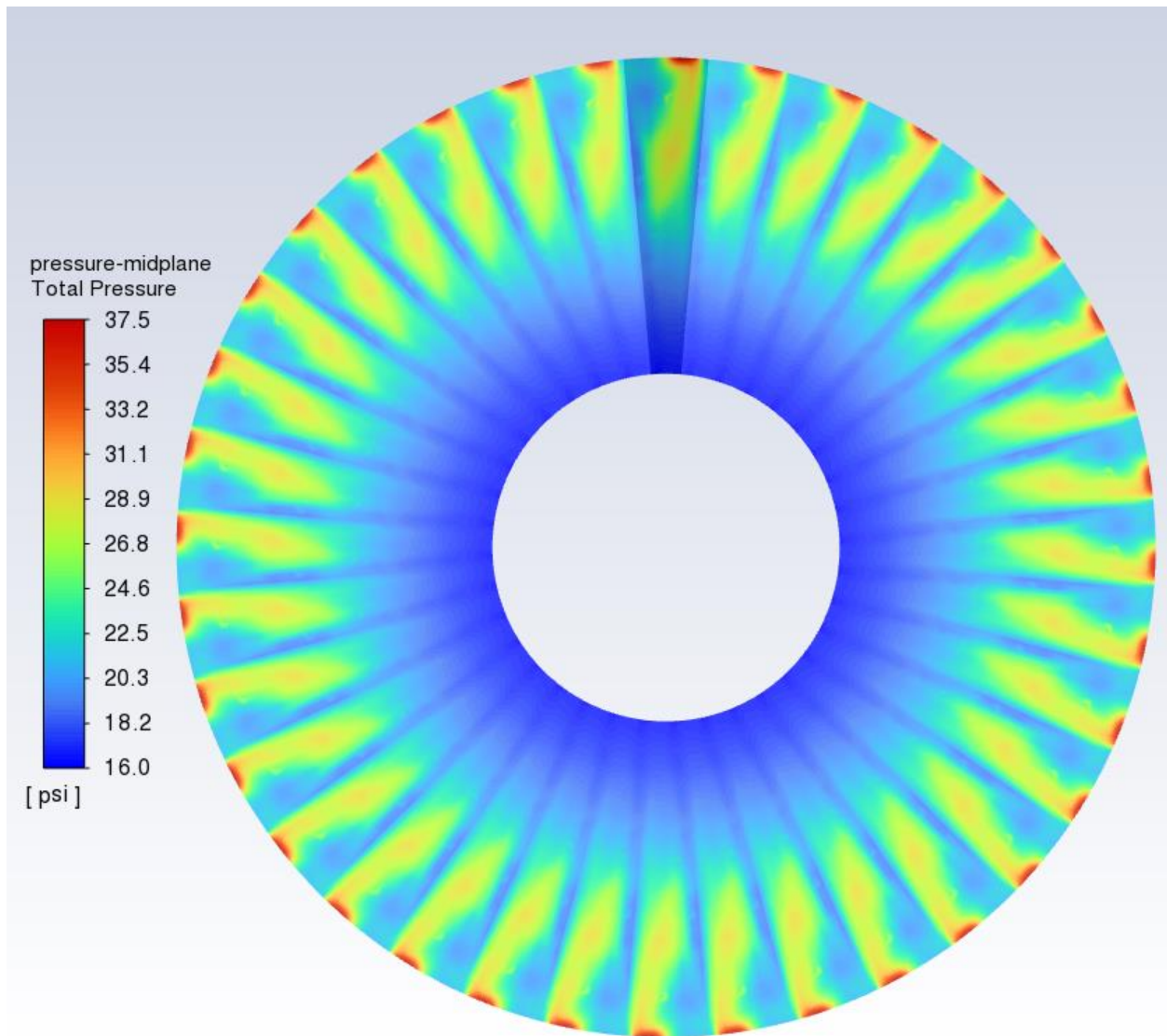


Figure 32. Pressure contour of Rotor and Stator Mid-Section Plane Stage 1 at Sea Level

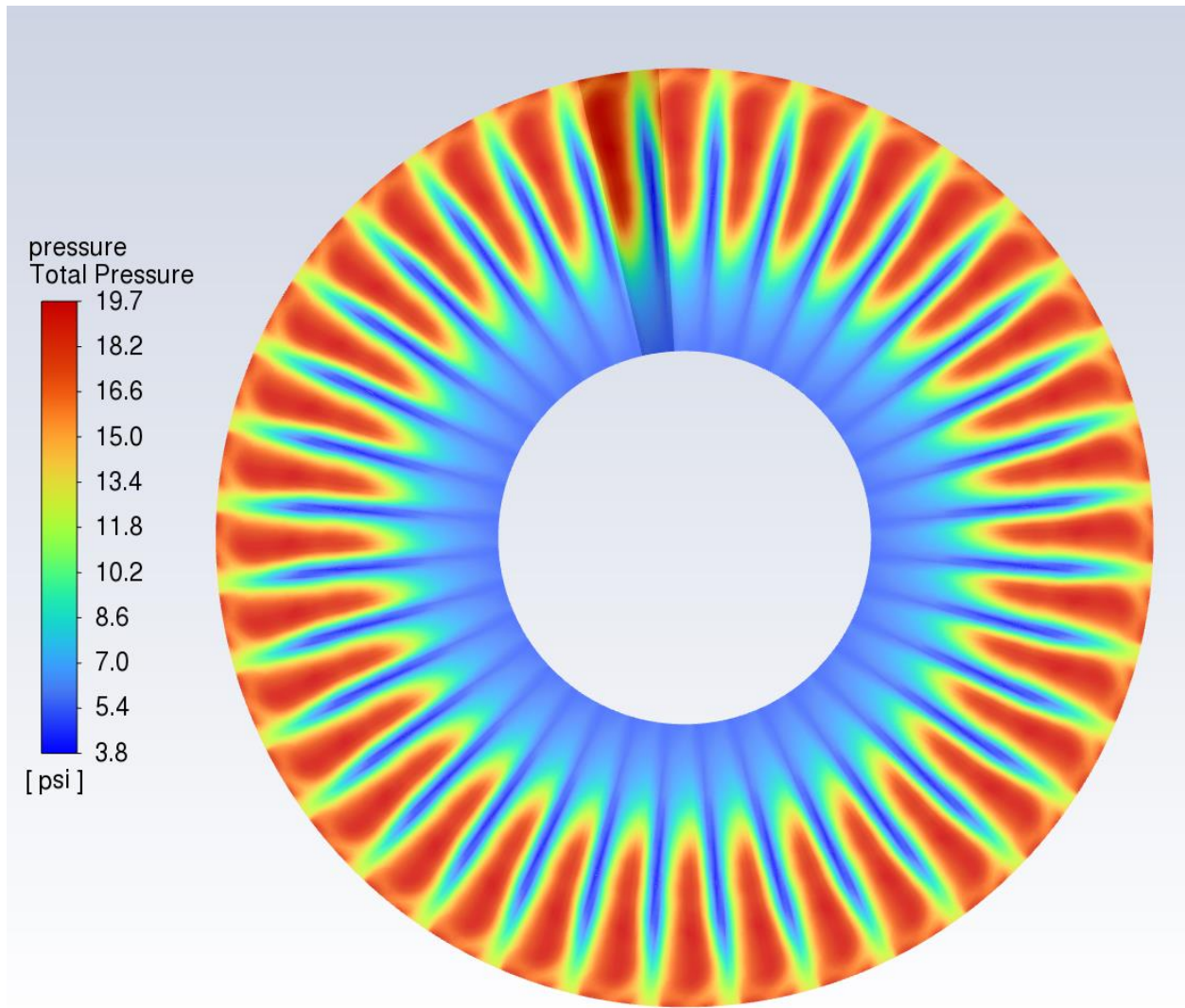


Figure 33. Pressure Contour of Rotor and Stator: Mid-section Plane (Stage 2 at 40,000 ft)

pressure-1
Total Pressure
129.8
118.0
106.3
94.6
82.8
71.1
59.4
47.6
35.9
24.2
12.4
[psi]

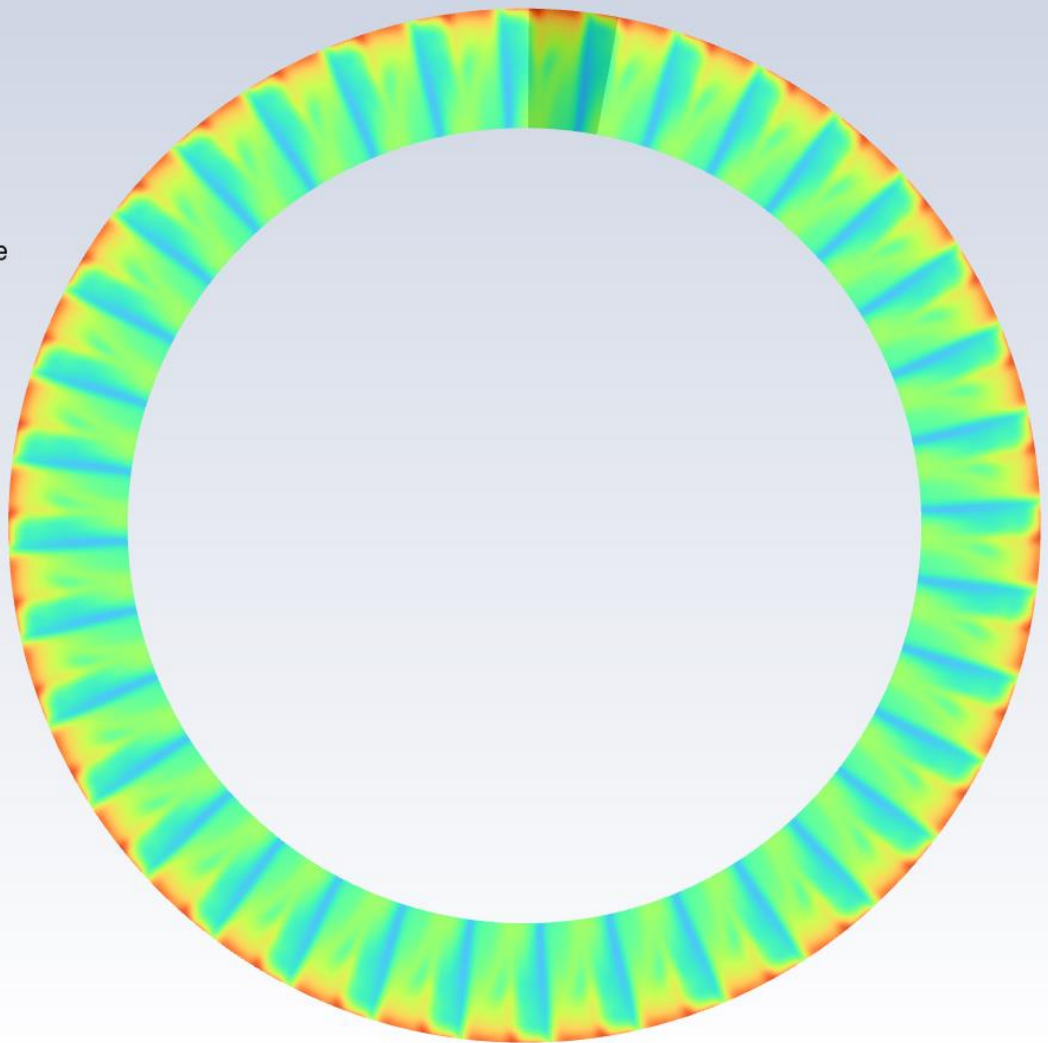


Figure 34. Pressure Contour of Rotor and Stator: Mid-section Plane (Stage 11 at 40,000 ft)

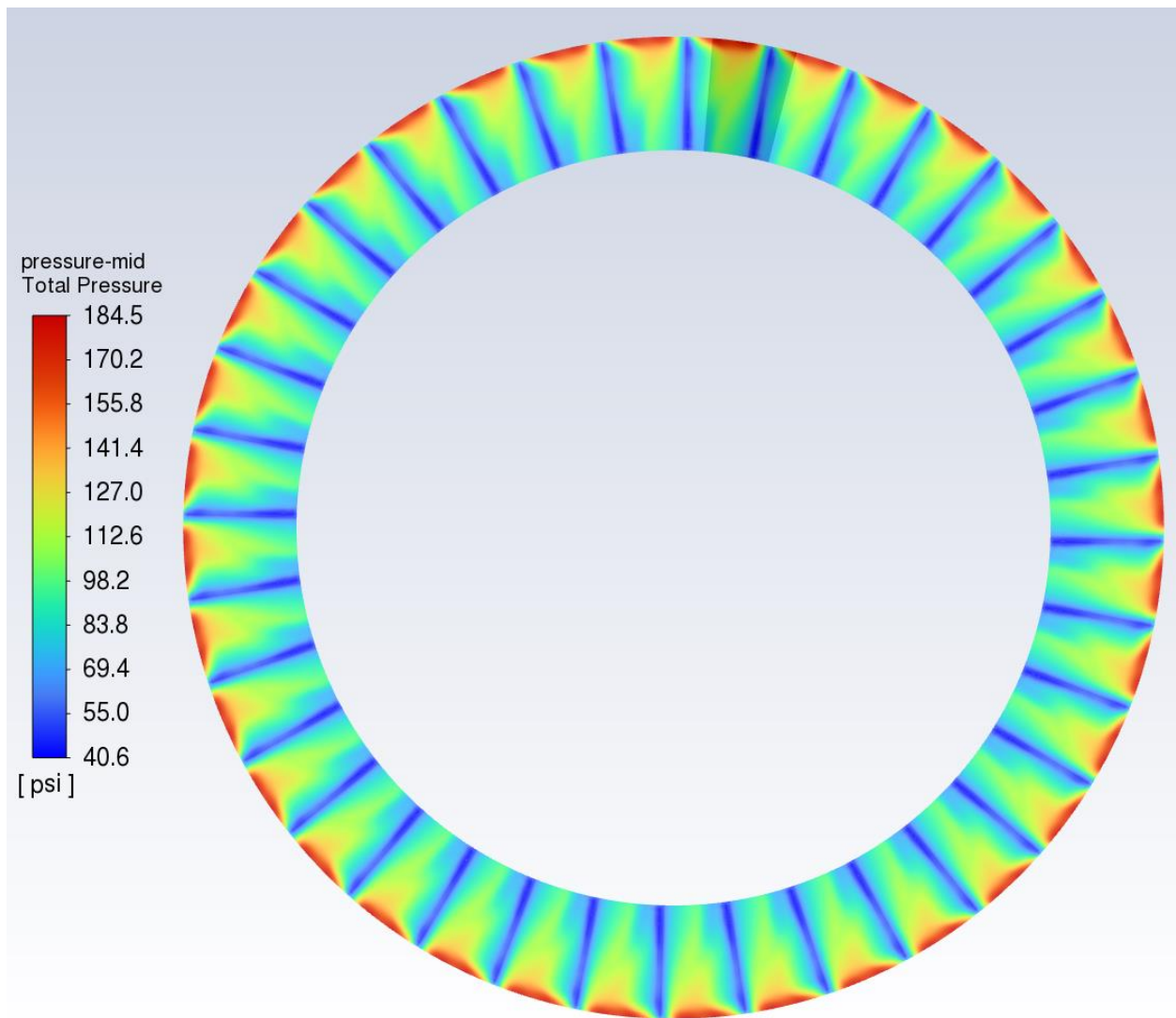


Figure 35. Pressure Contour of Rotor and Stator: Mid-section Plane (Stage 11 at Sea Level)

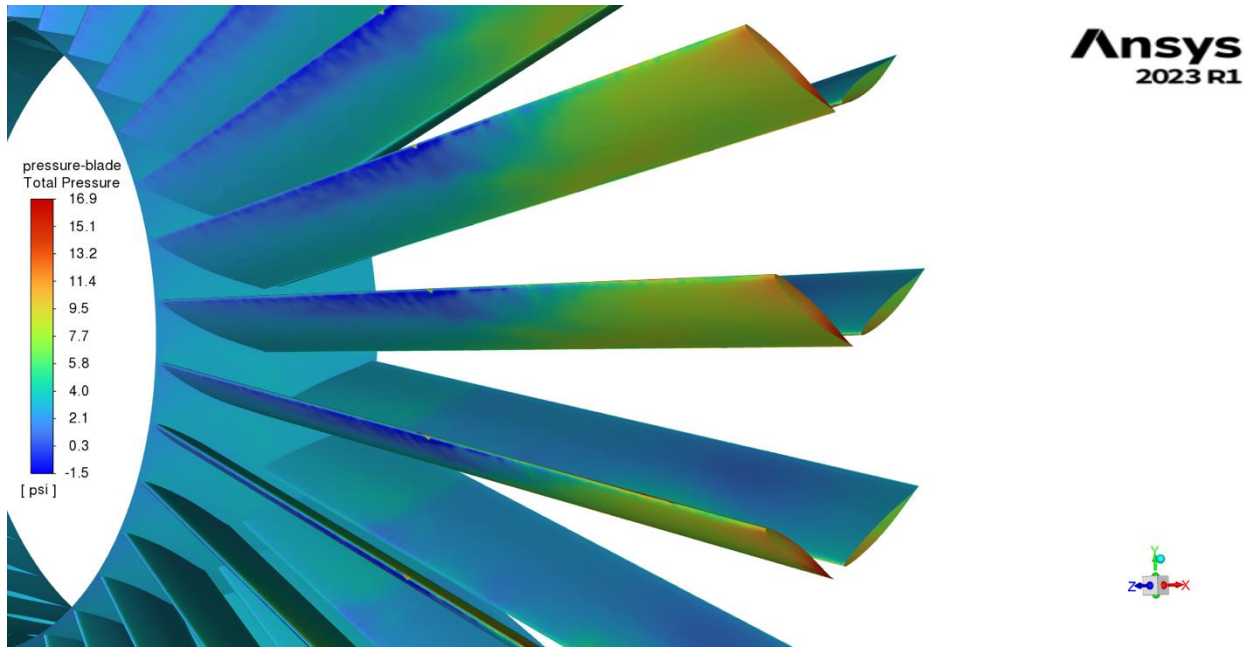


Figure 36. Pressure Contour of Rotor and Stator: Blades and Hub Walls (Stage 1 at 40,000 ft)

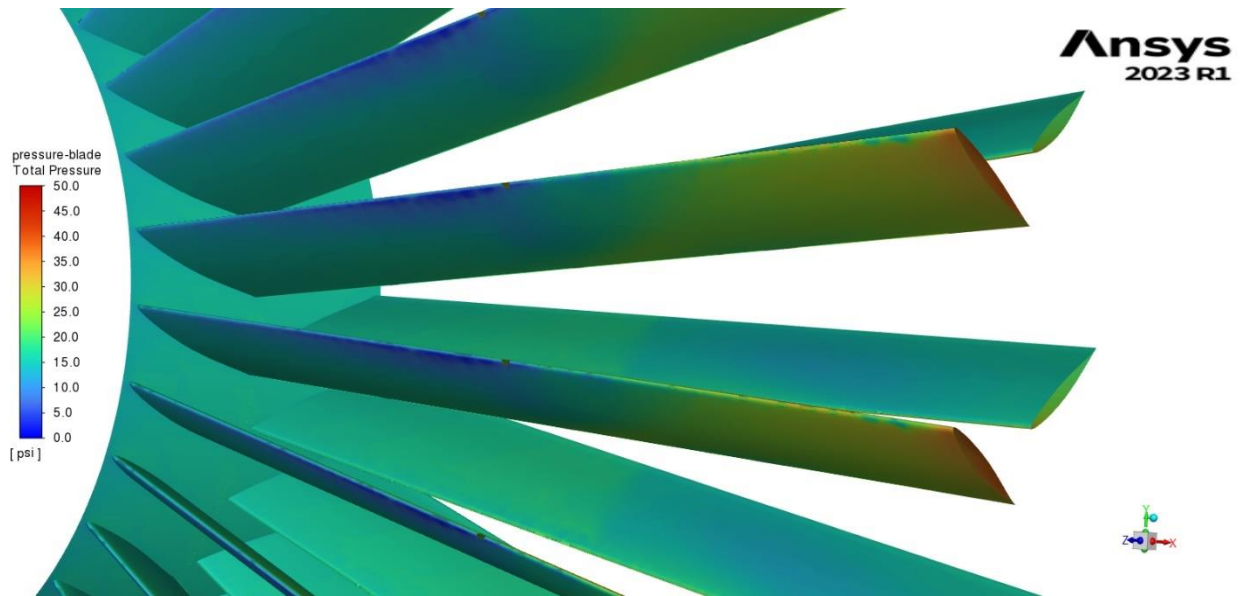


Figure 37. Pressure Contour of Rotor and Stator: Blades and Hub Walls (Stage 1 at Sea Level)

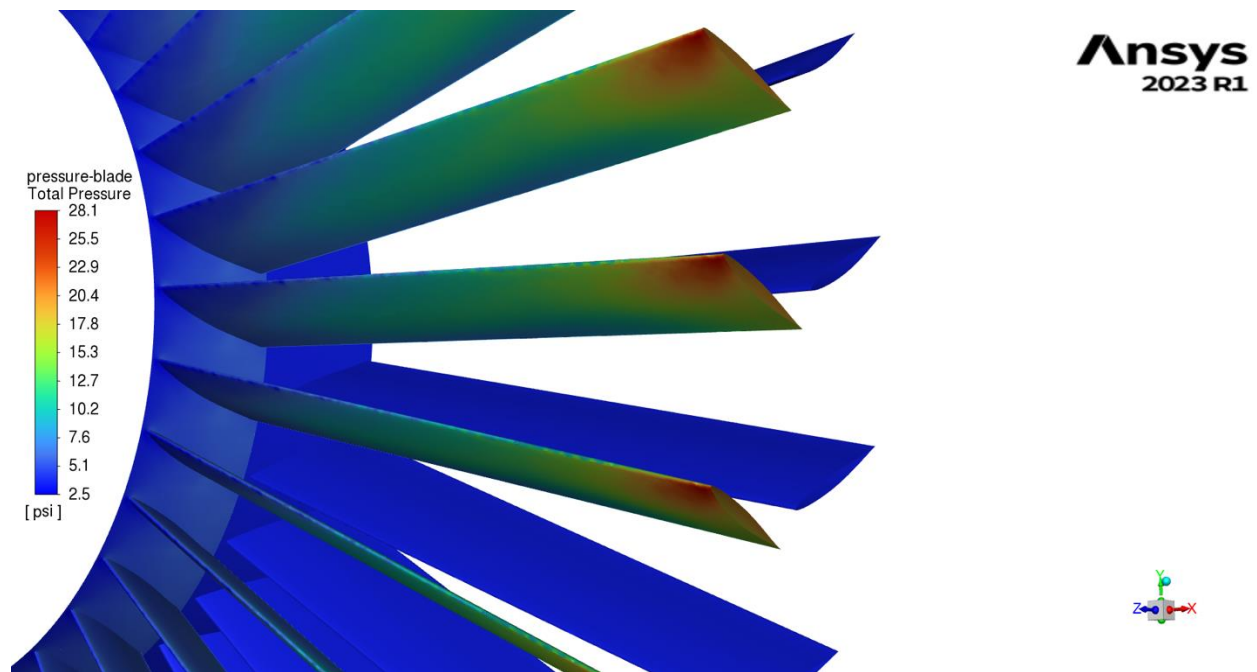


Figure 38. Pressure Contour of Rotor and Stator: Blades and Hub Walls (Stage 2 at 40,000 ft)

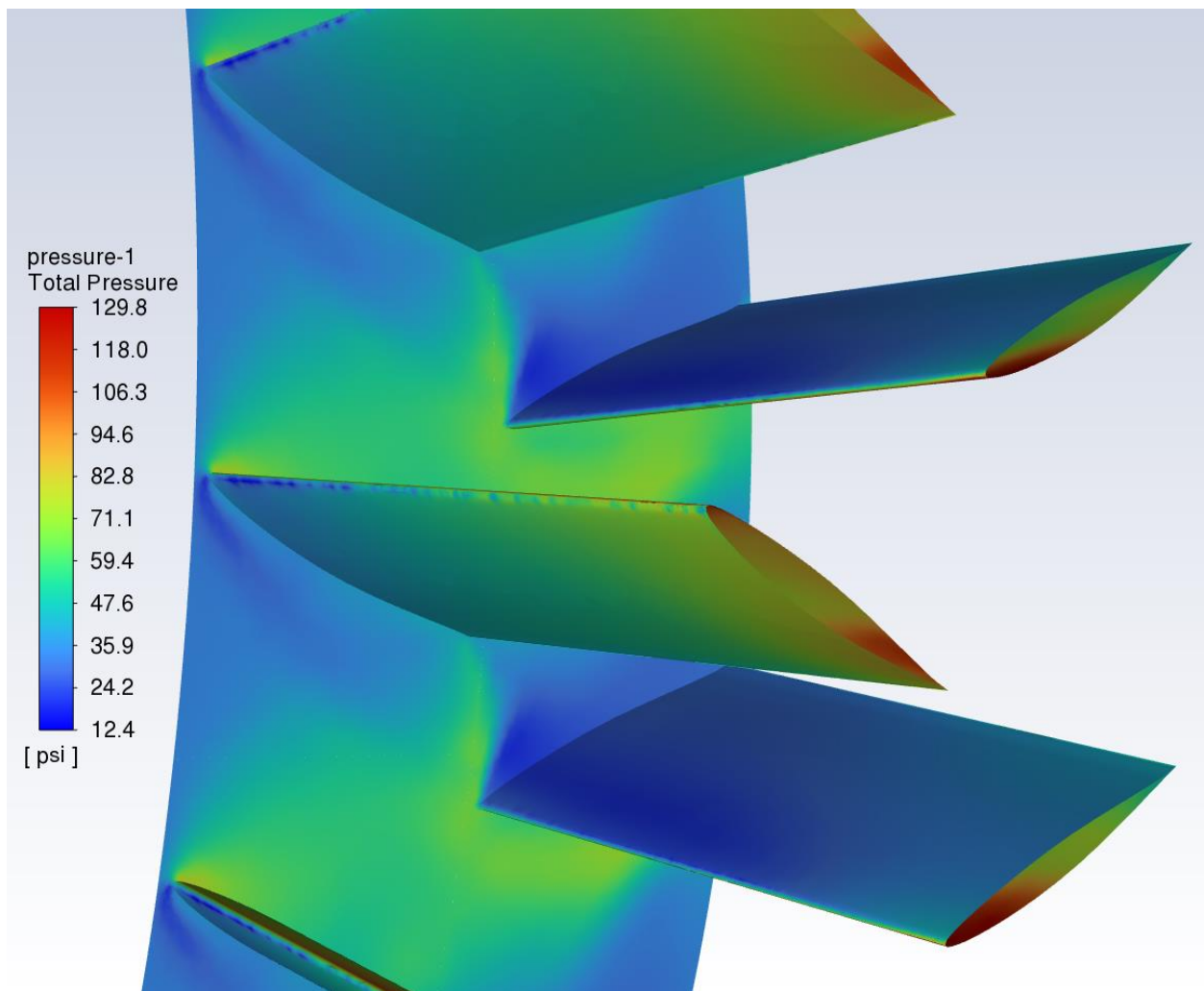


Figure 39. Pressure Contour of Rotor and Stator: Blades and Hub Walls (Stage 11 at 40,000 ft)

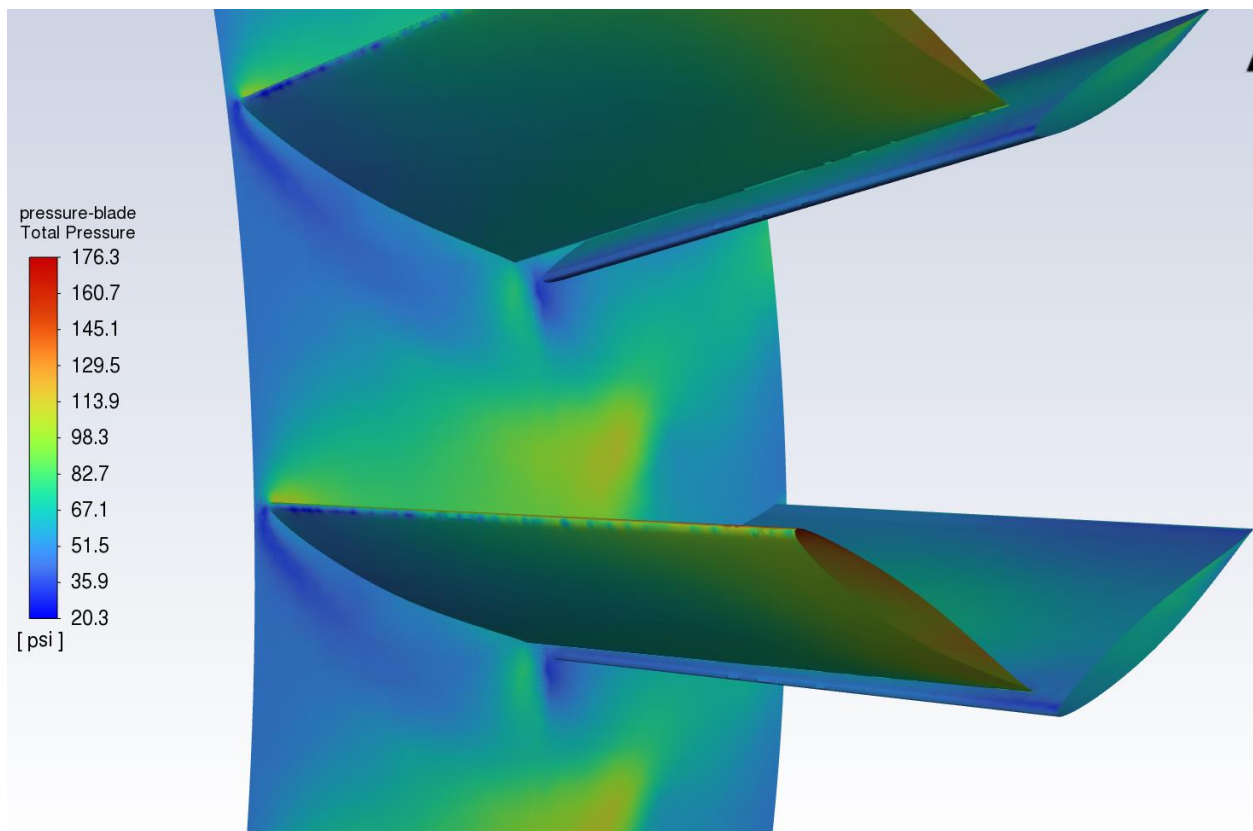
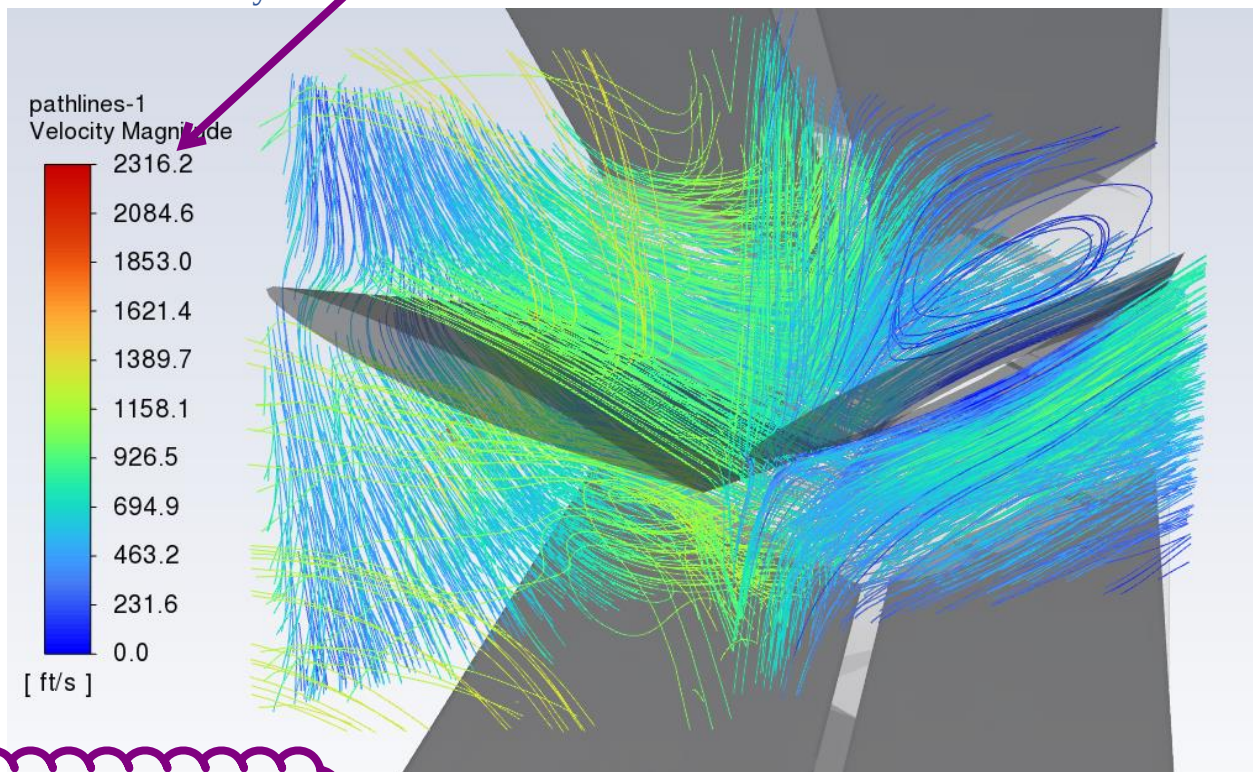


Figure 40. Pressure Contour of Rotor and Stator: Blades and Hub Walls (Stage 11 at Sea Level)

Very fast

8.4. Velocity Streamline Plots



Use thicker and fewer streamlines to make this more readable.

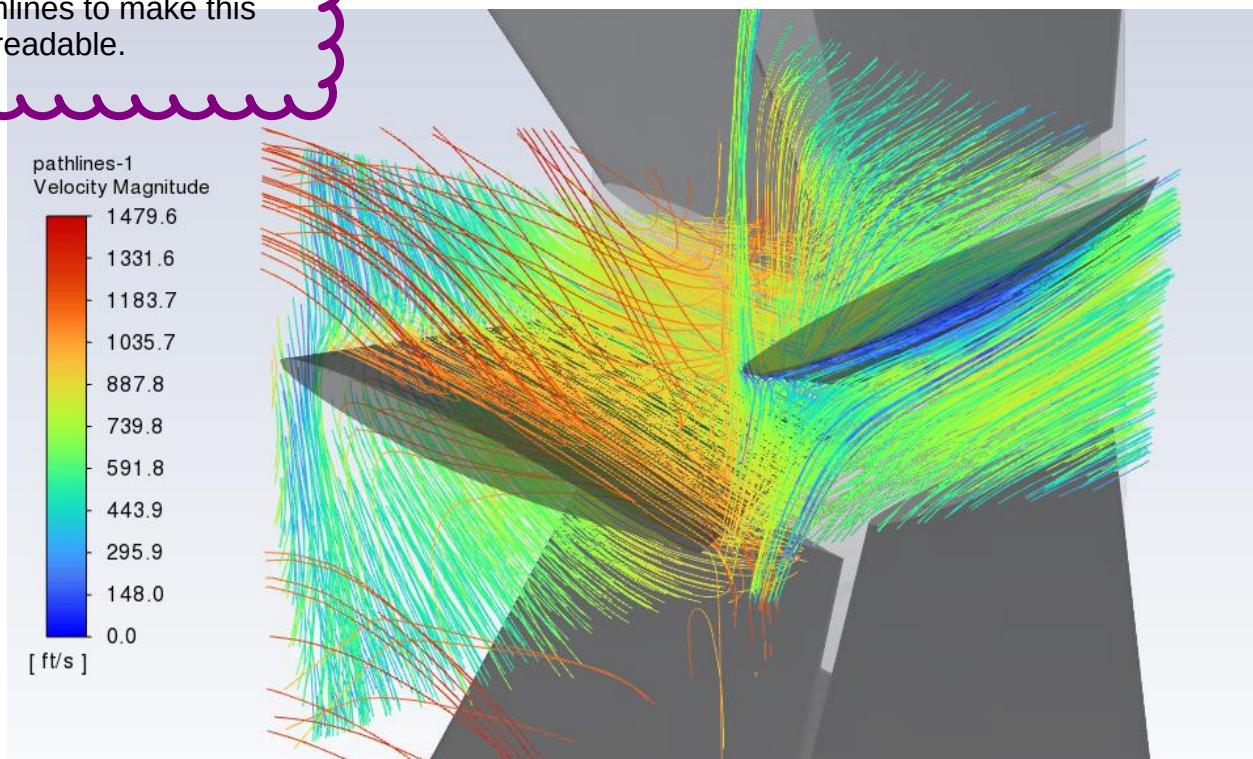


Figure 42. Velocity Streamline Plot Through Compressor Stage (Stage 1 at Sea Level)

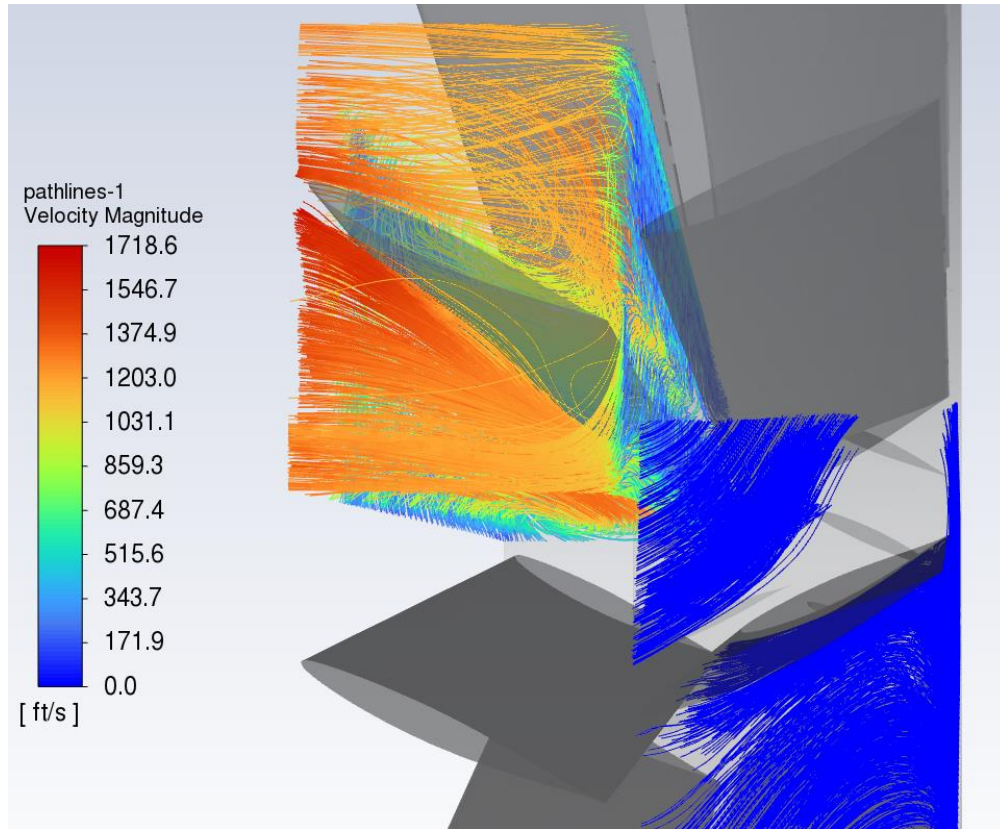


Figure 43. Velocity Streamline Plot Through Compressor Stage (Stage 2 at 40,000 ft)

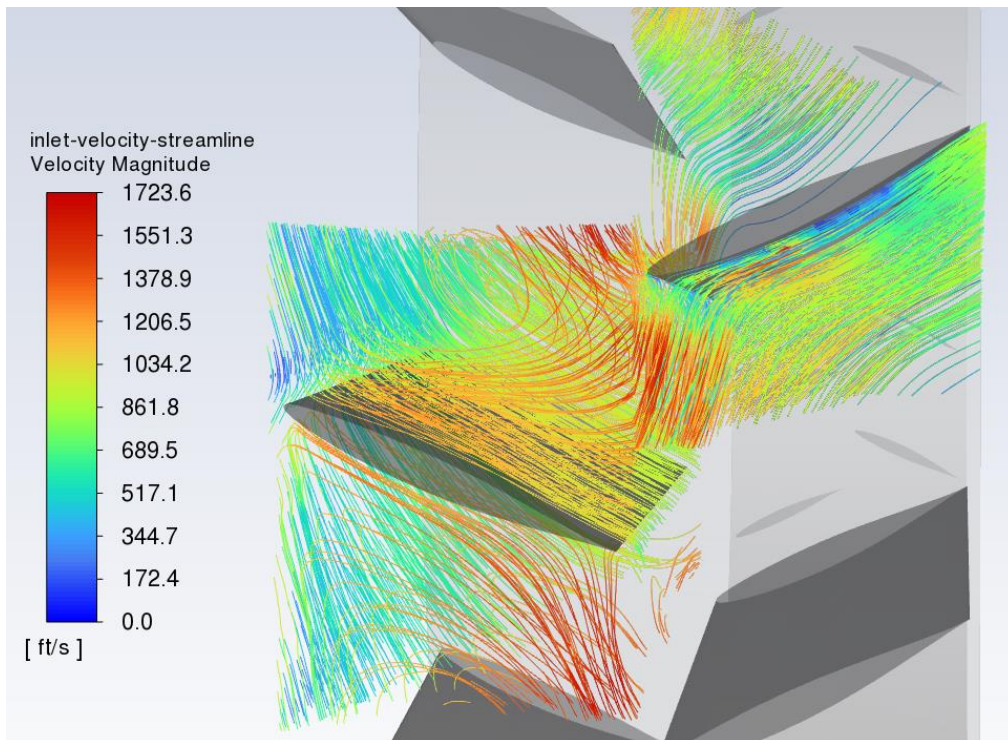


Figure 44. Velocity Streamline Plot Through Compressor Stage (Stage 11 at 40,000 ft)

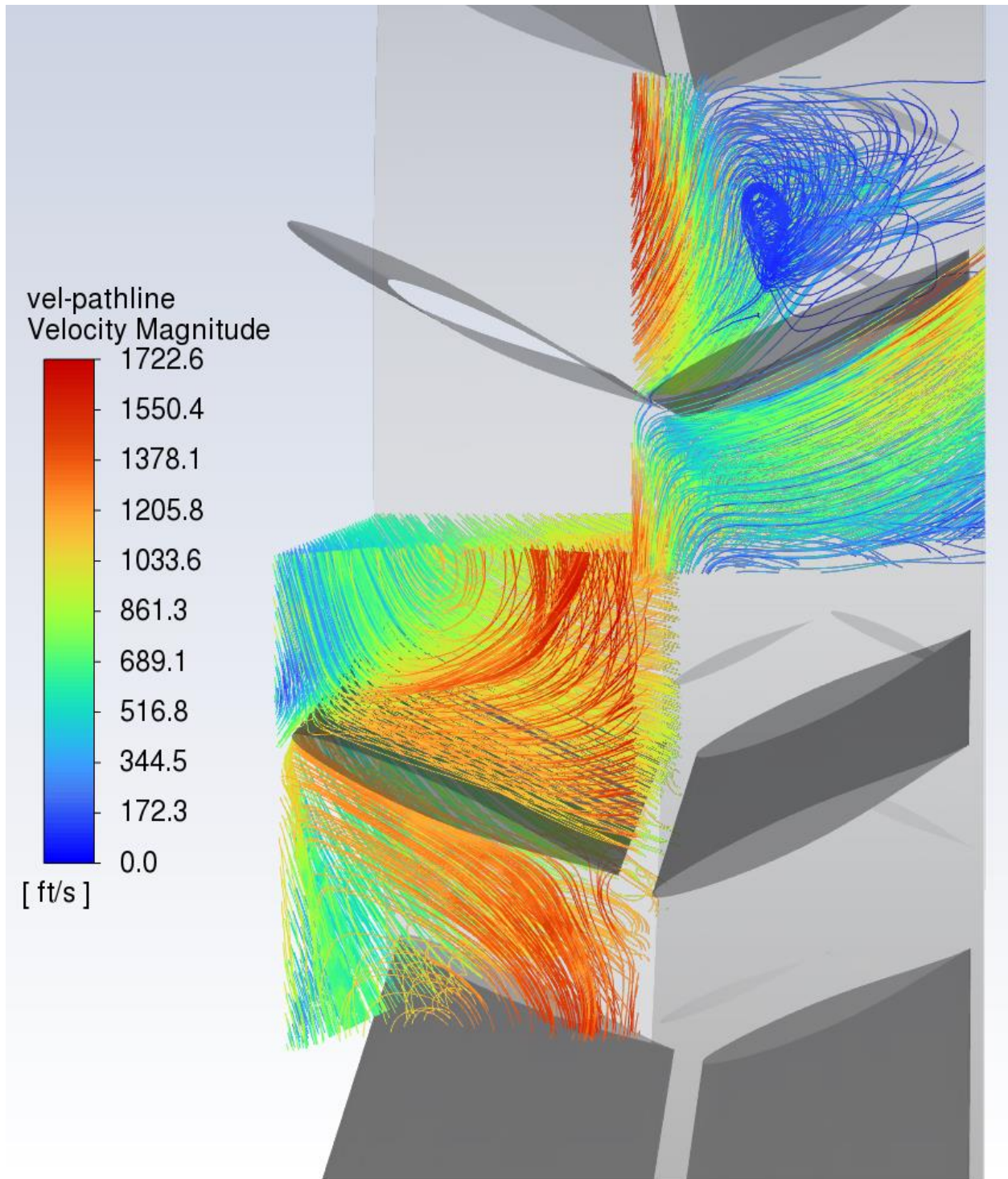


Figure 45. Velocity Streamline Plot Through Compressor Stage (Stage 11 at Sea Level)

8.5. Tabulated Results

Parameter	Value			
	Stage 1 at 40,000 ft	Stage 11 at 40,000 ft	Stage 1 at Sea Level	Stage 11 at Sea Level
Entrance temperature [°F]	-69.34	350.03	59.00	332.59
Exit temperature [°F]	5.66	442.57	120.26	447.39
Entrance Pressure [psi]	2.72	34.97	14.70	49.28
Exit Pressure [psi]	4.49	49.68	19.54	70.67
Pressure Increase [psi]	2.13	14.71	4.84	21.39
Compression ratio	1.65	1.42	1.33	1.43
Power usage [hp]	21.23	753.51	113.83	1180.93
Isentropic operating efficiency (%)	68.02	49.19	66.15	39.00

Table 13. Tabulated Results – Temperature, CPR, Power Usage, and Operating Efficiency

The power calculations should also be shown in the hand calculation section.

9. Discussion

To get the overall compression ratio of the entire axial compressor, interpolation was used to predict the rest of the stages compression ratio by using the stages that were simulated. As mentioned before, stages 1, 2, and 11 were selected to be simulated at both the design altitude of 40,000 ft. and at sea level. These stages were selected because they provide the most important data points for estimating the compression ratio of other stages.

Stages 1 and 2 can be used to see the overall trend of the compression ratio for the first 10 stages where the shroud wall is a constant diameter, and the hub of the compressor is increasing. The diameter of the hub increases at a constant angle of 5 degrees which means it is possible for the first two stages to predict these first 10 stages. The hand calcs as shown in appendix A show that the compression ratio of the first 10 stages hover around 1.2-2 for all the stages which means it is realistic to assume that the first two stages will be able to predict the first 10 stages.

Then, stage 11 is simulated to predict the rest of the stages from 11-18. In appendix A, stages 11-18 have a compression ratio of about 1.1 and stages 12-18 have similar compression ratios as stage 11. This means using the simulation results from stage 11 should be able to accurately extrapolate the remaining compression ratios of the stages. Then, stage 11 is simulated to predict the rest of the stages from 11-18.

However, there were some issues with stage 2 at 40k ft. The interface between the rotor and stator was not set up correctly leading to an inaccurate simulation as the exit of the rotor did not properly enter the rotor. Additionally, the operating pressure was not properly set for stage 2, leading to an incorrect pressure result. This means that to extrapolate for stages 1-10, only the results from stage 1 are used.

Using the Tabulated Results from section 8.5. Stage 1 had a compression ratio of 1.65 and stage 11 had a compression ratio of 1.42. Extrapolating the stage 2-9 and stage 12-18 is done by assuming that their compression ratios were proportional to the outlet temperature and pressures. This provides a more accurate extrapolation than using a linear line or assuming a constant compression ratio. Below is a table showing the extrapolation done using the results from stage 1 and stage 11. It was found that only 14 stages would be required to reach the desired compression ratio of 22:1 at the operating altitude of 40,000 ft.

Stage	Inlet temp [oF]	Outlet temp [oF]	Inlet pressure [psi]	Outlet pressure [psi]	CPR	L_blade_rotor [in]
1	-69.34	5.66	2.72	4.49	1.649	21.69
2	5.66	95.53	4.49	8.32	1.855	20.237
3	95.53	123.04	8.32	9.86	1.184	18.826
4	123.04	153.46	9.86	11.78	1.195	17.415
5	153.46	186.00	11.78	14.12	1.198	16.005
6	186.00	219.85	14.12	16.88	1.196	14.594
7	219.85	254.19	16.88	20.06	1.188	13.184
8	254.19	288.23	20.06	23.61	1.177	11.773
9	288.23	320.29	23.61	27.35	1.158	9.657
10	320.29	350.03	27.35	31.18	1.140	8.246
11	350.03	379.06	34.97	49.68	1.421	7.894
12	379.06	406.54	49.68	55.61	1.119	7.194
13	406.54	432.27	55.61	61.61	1.108	6.494
14	432.27	456.04	61.61	67.55	1.096	5.794

Table 14. Interpolation Values for Stages 1-14 at 40,000 ft. Altitude (Highlighted values represent simulation values)

Table 15 shows the comparison of the results from the CFD simulation to the hand calculation values.

Parameter	Value			
	Stage 1 at 40,000 ft	Stage 11 at 40,000 ft	Stage 1 at Sea Level	Stage 11 at Sea Level
Exit temperature [°F]	5.66 (34.11)	442.57 (379.06)	120.26 (162.45)	447.39 (361.52)
Exit Pressure [psi]	4.49 (6.19)	49.68 (39.56)	19.54 (21.14)	70.67 (55.90)
Compression ratio	1.65 (2.28)	1.42 (1.13)	1.33 (1.44)	1.43 (1.13)

Table 1515. Comparison of Simulation Results to Hand Calculation Values

Note: values in parentheses are from hand calculation.

Watch significant figures

10. Conclusion

The design has benefits of being modular and scalable. More or fewer stages can be added or subtracted very easily to change the desired compression ratio, as was necessary following the results of the simulations which indicated only 14 stages were needed to achieve the design compression ratio of 22:1. Additionally, the design uses a constant profile airfoil which greatly simplifies construction, calculations, and simulations. These are important factors because simplifying these aspects of designs means it will likely be more accurately designed which means the axial compressor will behave more like what was predicted.

A notable feature of the design is that the first 10 stages have the hub diameter increase while the later stages 11-18 have a constant hub diameter with a decreasing wall shroud diameter. This design was chosen because it was planned to run at higher rotational speeds for some of the stages. Decreasing the shroud diameter would mean that the tip speeds of the blades would also decrease. However, it was ultimately determined to run all stages at 4,000 rpm so this design feature was not fully utilized. In the future this aspect of the design could be explored more to analyze whether the shroud should be tapered at all, or the hub diameter should be the only thing increasing.

More future work can be done by analyzing and simulating more of the stages for a more accurate interpolation of the compression ratio of the stages. Stages 1, 2, and 11 are simulated which provide an adequate amount of data points to predict the trends of the compressor. However, simulating more stages would give an even more accurate prediction of the compression ratio.

Other work can be done to explore a changing airfoil profile for the compressor blades to reflect that of real-life axial compressors. It makes sense to use a changing profile because each section of the airfoil will experience a different speed as the blade moves from the hub (a slower speed) towards the shroud wall (a faster speed).

These future improvements may lead to a design that can run at slower speeds, use fewer stages, require less power, and can be a smaller size.

11. Design Time Estimate

The total design time is estimated to be 160 hours, including conducting the background research on the compressor design, calculating estimated pressure and temperature drop, modeling, meshing, and simulating the compressor stage.

Section	Man Hours (hr)
Background Research	5
Hand Calcs	15
Modeling	30
Meshing	40
Simulation	70
Total	160

12. References

Wright, L. C. *Blade Selection for a Modern Axial-Flow Compressor*. Vol. 304, 1 Jan. 1974, p. 603.

13. Appendix A: Hand-calculated Temperature, Pressure, and CPR

At 40,000 ft:

Stage	Entrance temp [°F]	Exit temp [°F]	Entrance pressure [psi]	Exit pressure [psi]	CPR
1	-69.34	34.11	2.72	6.19	2.277
2	34.11	95.53	6.19	9.34	1.507
3	95.53	123.04	9.34	11.06	1.184
4	123.04	153.46	11.06	13.21	1.195
5	153.46	186.00	13.21	15.84	1.198
6	186.00	219.85	15.84	18.94	1.196
7	219.85	254.19	18.94	22.50	1.188
8	254.19	288.23	22.50	26.49	1.177
9	288.23	320.29	26.49	30.68	1.158
10	320.29	350.03	30.68	34.97	1.140
11	350.03	379.06	34.97	39.56	1.131
12	379.06	406.54	39.56	44.29	1.119
13	406.54	432.27	44.29	49.06	1.108
14	432.27	456.04	49.06	53.80	1.096
15	456.04	477.67	53.80	58.38	1.085
16	477.67	496.95	58.38	62.69	1.074

At Sea Level:

Stage	Entrance temp [°F]	Exit temp [°F]	Entrance pressure [psi]	Exit pressure [psi]	CPR
1	59	162.45	14.70	21.14	1.439
2	162.45	187.01	21.14	24.21	1.145
3	187.01	214.51	24.21	28.01	1.157
4	214.51	244.94	28.01	32.69	1.167
5	244.94	277.48	32.69	38.29	1.171
6	277.48	311.32	38.29	44.80	1.170
7	311.32	345.67	44.80	52.18	1.165
8	345.67	379.70	52.18	60.32	1.156
9	379.70	411.77	60.32	68.78	1.140
10	411.77	441.51	68.78	77.35	1.125
11	441.51	470.54	77.35	86.42	1.117
12	470.54	498.02	86.42	95.70	1.107
13	498.02	523.74	95.70	105.00	1.097
14	523.74	547.52	105.00	114.16	1.087
15	547.52	569.14	114.16	122.97	1.077
16	569.14	588.42	122.97	131.22	1.067
17	588.42	605.15	131.22	138.70	1.057
18	605.15	619.14	138.70	145.18	1.047
19	619.14	630.18	145.18	150.45	1.036
20	630.18	638.08	150.45	154.30	1.026

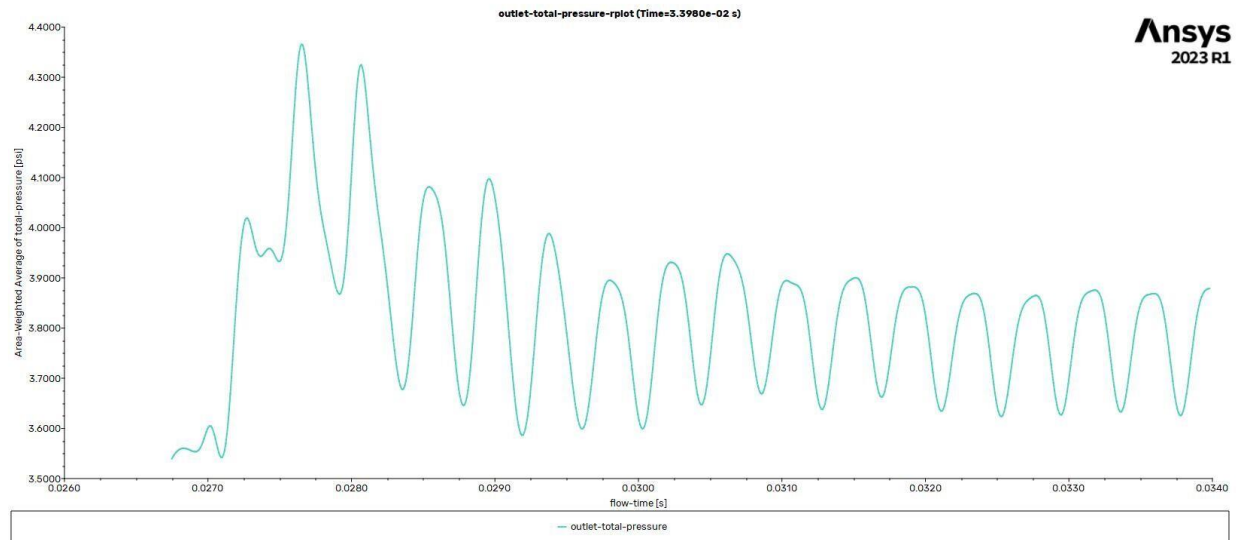
14. Appendix B: Blade Length at Each Stage

Stage	Length of blade [in]
1	21.69
2	20.237
3	18.826
4	17.415
5	16.005
6	14.594
7	13.184
8	11.773
9	9.657
10	8.246
11	7.894
12	7.194
13	6.494
14	5.794
15	5.094
16	4.394
17	3.694
18	2.994
19	2.294
20	1.594

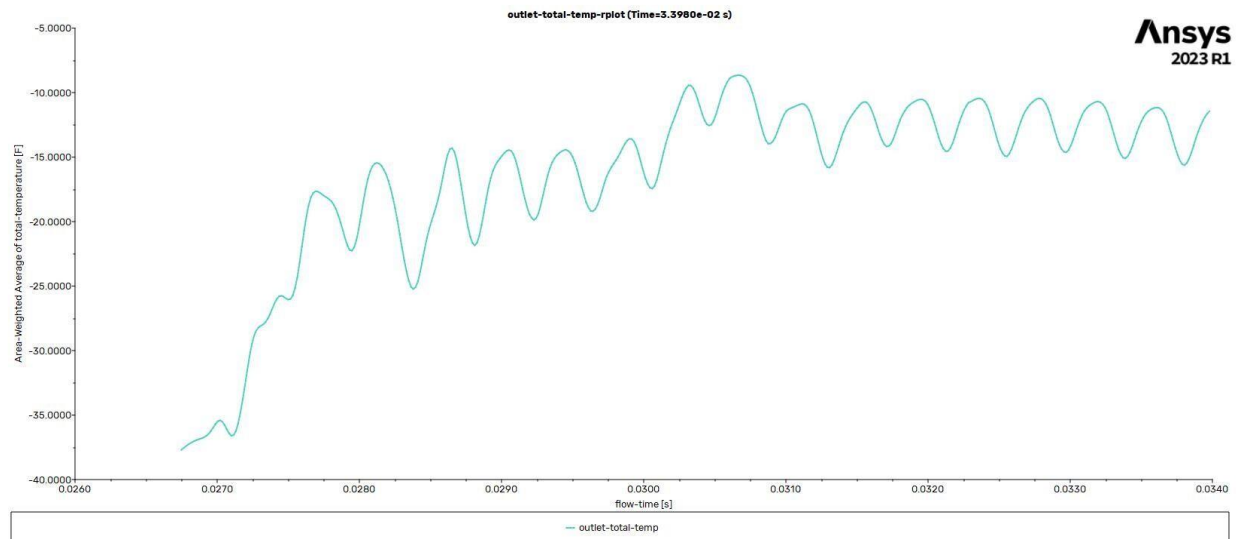
15. Appendix C: Interpolated Results for Sea Level

Stage	Inlet temp [oF]	Outlet temp [oF]	Inlet pressure [psi]	Outlet pressure [psi]	CPR	l_blade_rotor [in]
1	59	120.26	14.70	19.54	1.250	21.69
2	120.26	187.01	19.54	28.61	1.464	20.237
3	187.01	214.51	28.61	33.10	1.157	18.826
4	214.51	244.94	33.10	38.63	1.167	17.415
5	244.94	277.48	38.63	45.24	1.171	16.005
6	277.48	311.32	45.24	52.94	1.170	14.594
7	311.32	345.67	52.94	61.66	1.165	13.184
8	345.67	379.70	61.66	71.28	1.156	11.773
9	379.70	411.77	71.28	81.27	1.140	9.657
10	411.77	441.51	81.27	91.40	1.125	8.246
11	332.59	447.39	49.28	70.67	1.606	7.894
12	447.39	474.87	70.67	78.45	1.110	7.194
13	474.87	500.59	78.45	86.27	1.100	6.494
14	500.59	524.37	86.27	93.98	1.089	5.794
15	524.37	546.00	93.98	101.41	1.079	5.094
16	546.00	565.27	101.41	108.38	1.069	4.394
17	565.27	582.01	108.38	114.70	1.058	3.694
18	582.01	595.99	114.70	120.18	1.048	2.994
19	595.99	607.03	120.18	124.64	1.037	2.294
20	607.03	614.93	124.64	127.90	1.026	1.594
21	614.93	614.93	127.90	127.90	1.000	0.894
22	614.93	614.93	127.90	127.90	1.000	0.194
				Total CPR	16.995	

16. Appendix D: Pressure and Temperature Monitors for Stage 1: 40,000'



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